

Design Of Ultra-low-distortion Band-pass Filters

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- Sinusoids for ADC histogram and THD testing: $\text{THD} \leq -140 \text{ dBc}$
- Work done with Texas Instruments India

- Sangeeta Kumar, Rajashekar Goroju, Dileep Bhat, K. S. Rakshitdatta and Nagendra Krishnapura, “Design Considerations for Low-Distortion Filter and Oscillator ICs for Testing High-Resolution ADCs,” *TCAS I*, Sept. 2019.
- Subha Sarkar, Rajat Agarwal, and Nagendra Krishnapura, “Bandpass Filter and Oscillator ICs with $\text{THD} < -140 \text{ dBc}$ at 10 Vppd for Testing High-Resolution ADCs,” *ISSCC 2023*.

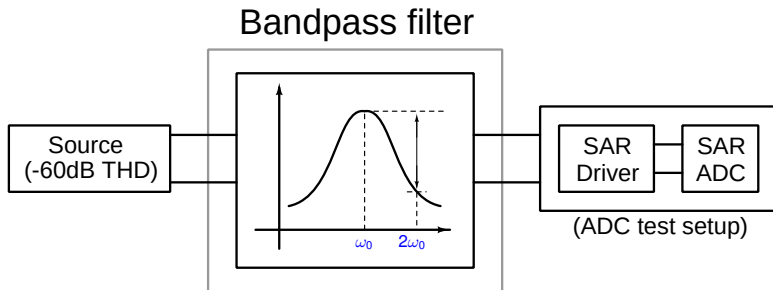
Low-distortion Active Band-pass Filters

- Low distortion sinusoidal source for testing 16-20 bit ADCs
- Low cost compared to bench-top sources
- On-chip solution that
 - Allows large scale testing
 - Can be integrated on board with the DUT

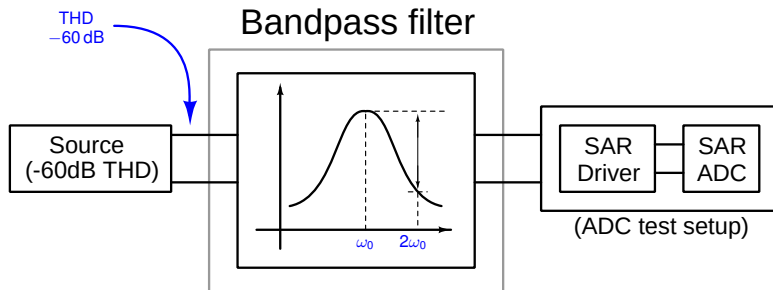


- Requirements from source:
 - Low distortion
 - Low frequency and amplitude drift
- Signal source specifications:
 - Signal amplitude: 10 V_{ppd} differential
 - Supply: 5.6 V
 - Frequencies: 1 kHz and 10 kHz
 - THD $\sim -120\text{ dB}$ for both frequencies
- Approaches
 - Bandpass filter: Filter out the harmonics of distorted source
 - Oscillator: Generates a low-distortion sine-wave

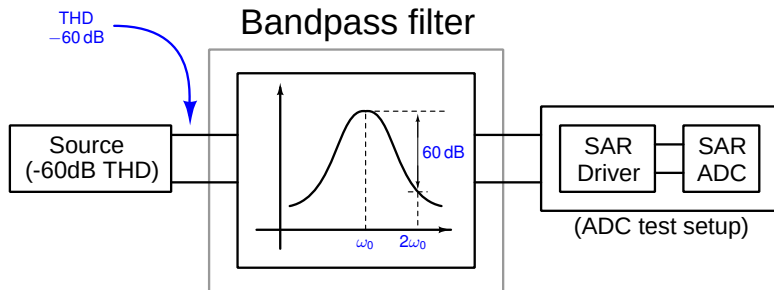
- Filter architecture
- Distortion analysis
- Opamp design
- Simulation and measurement results



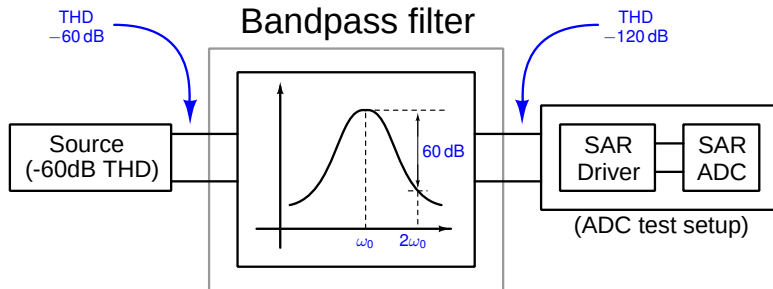
Bandpass filter requirements



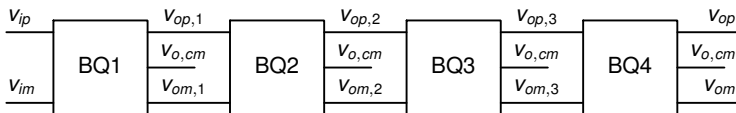
Bandpass filter requirements



Bandpass filter requirements

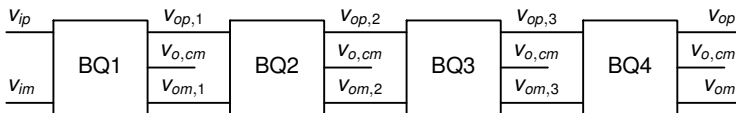


Filter architecture: Top level

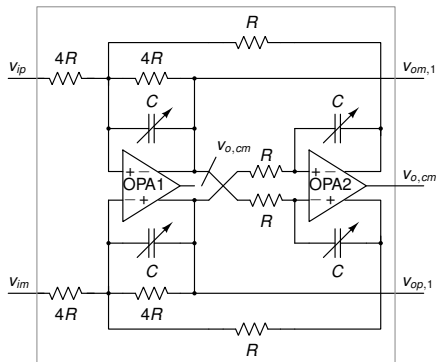


4 stages: Each $Q = 4$ for 60 dB attenuation at $2\omega_0$

Filter architecture: Top level



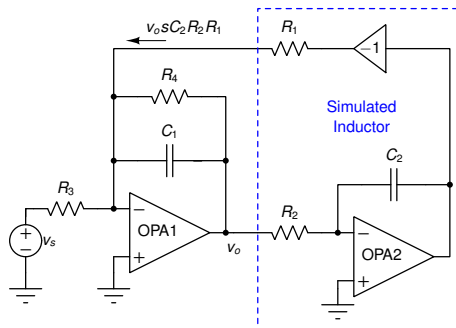
4 stages: Each $Q = 4$ for 60 dB attenuation at $2\omega_0$



Differential Tow-Thomas biquad, $Q=4$

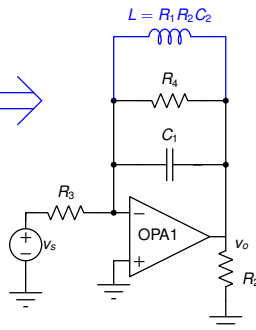
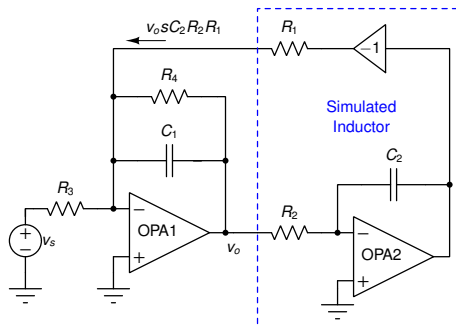
$R = 1.59 \text{ M}\Omega$ for 1 kHz and $159 \text{ k}\Omega$ for 10 kHz, $C = 100 \text{ pF}$

Filter architecture: Tow-Thomas biquad

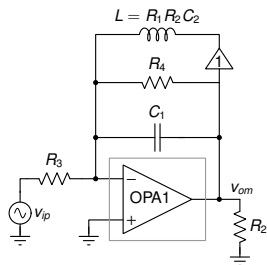


$$\frac{v_o}{v_s} = H_0 \frac{s/\omega_0 Q}{1 + s/\omega_0 Q + s^2/\omega_0^2}; \quad \omega_0 = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}; \quad Q = \frac{R_4}{\sqrt{R_1 R_2}} \sqrt{\frac{C_1}{C_2}}; \quad H_0 = \frac{R_4}{R_3}$$

Filter architecture: Tow-Thomas biquad

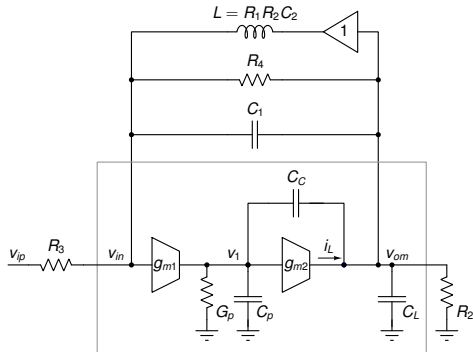
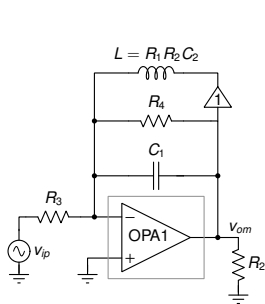


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- Miller-compensated opamp

Distortion due to output stage

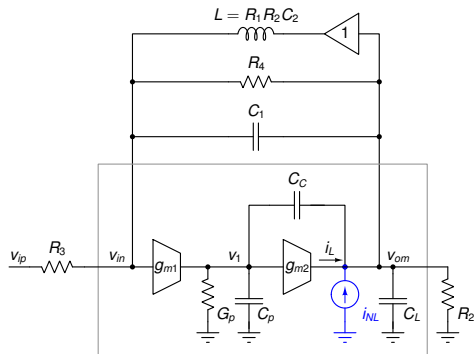
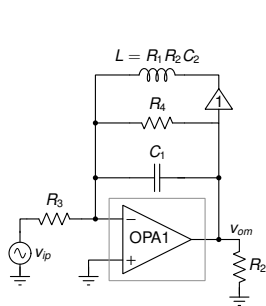


- Miller-compensated opamp

- g_{m1} : Differential pair
- g_{m2} : Common-source amplifier

- $$i_L = \underbrace{g_{(1,0)} v_1 + g_{(0,1)} v_{om}}_{\text{First Order}}$$

Distortion due to output stage

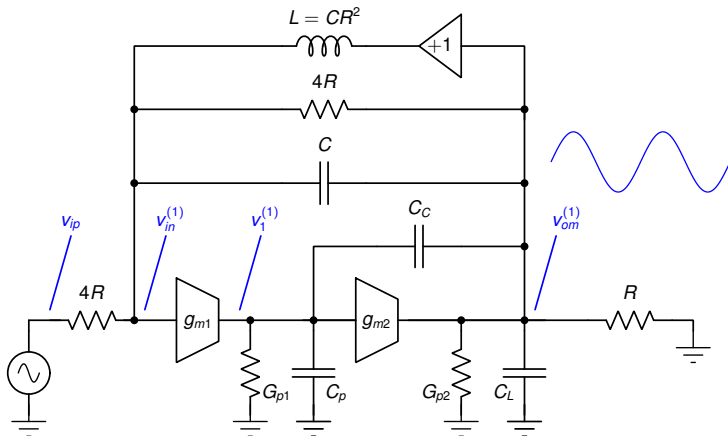


- Miller-compensated opamp

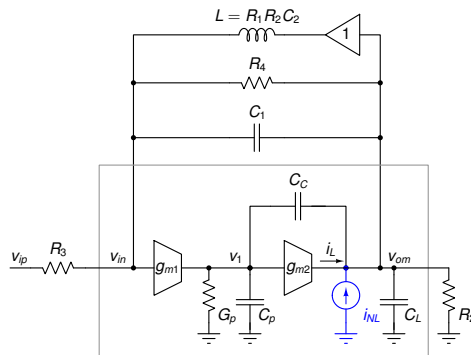
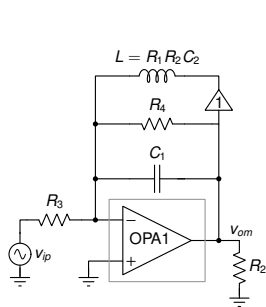
- g_{m1} : Differential pair
- g_{m2} : Common-source amplifier

- $$i_L = \underbrace{g_{(1,0)} v_1 + g_{(0,1)} v_{om}}_{\text{First Order}} + \underbrace{g_{(2,0)} v_1^2 + g_{(1,1)} v_1 v_{om} + g_{(0,2)} v_{om}^2}_{\text{Second Order}} + \dots$$

Distortion analysis: Linear circuit analysis



Distortion due to output stage

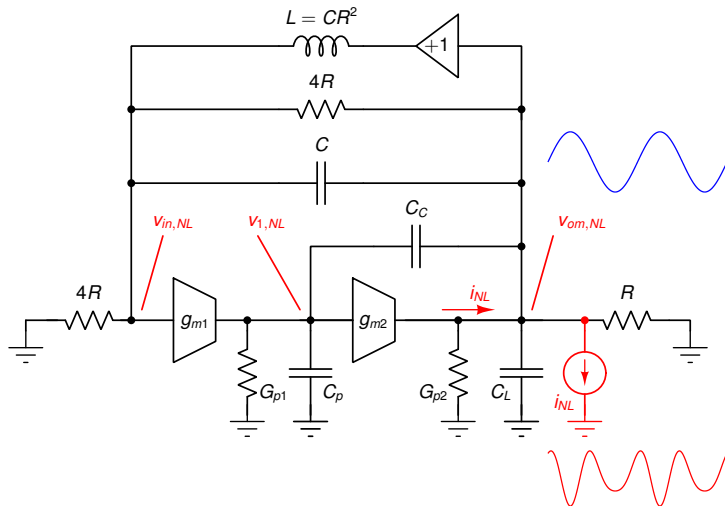


- Miller-compensated opamp

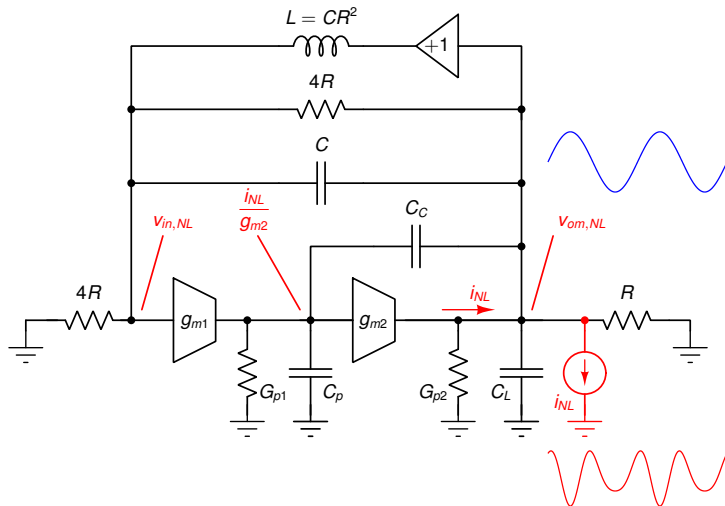
- g_{m1} : Differential pair
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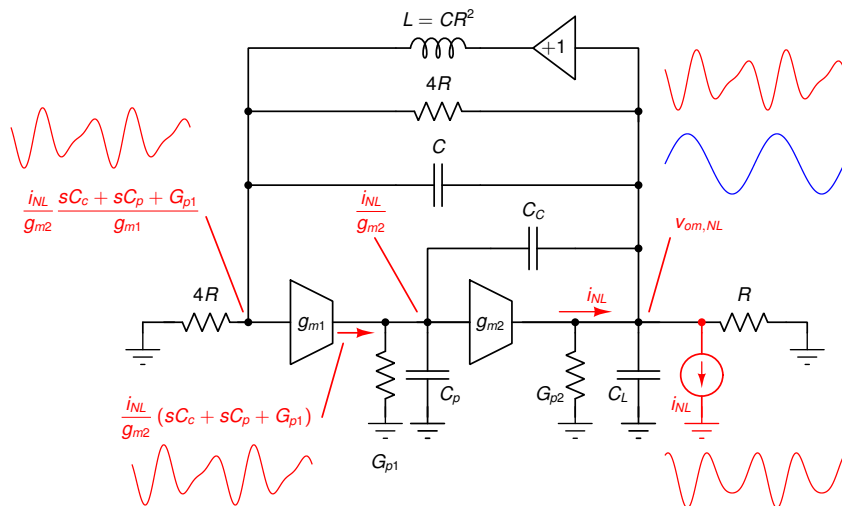
Distortion analysis: Nonlinear current injection



Distortion analysis: Nonlinear current injection



Distortion analysis: Nonlinear current injection



$$v_{om,NL} \approx \underbrace{-\frac{i_{NL}}{g_{m2}}}_{v_1} \underbrace{\frac{s(C_C + C_{p1}) + G_{p1}}{g_{m1}}}_{H_o(s)}$$

$H_o(s)$ is v_{om}/v_1 when i_{NL} is applied at the output of the opamp
Hence, output distortion depends on:

- i_{NL}/g_{m2} :
 - Decided by the operating region of output stage and load conditions
 - Absolute values are dependent on the model
- $H_o(s)$:
 - Dependent on the architecture of the opamp used
 - Independent of opamp scaling

Reduce $H_o(s)$

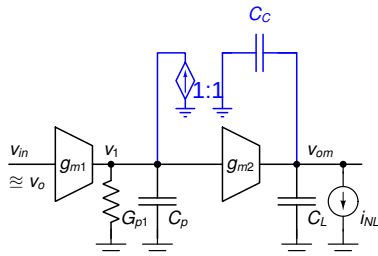
$$H_o(s)_{\text{Miller}} = \frac{s(C_C + C_p) + G_{p1}}{g_{m1}}$$

Bias first stage in weak inversion to increase g_{m1}/G_{p1}

$$H_o(s)_{\text{Miller}} = \frac{s(C_C + C_p) + G_{p1}}{g_{m1}}$$

Bias first stage in weak inversion to increase g_{m1}/G_{p1}

Indirect Miller Compensation

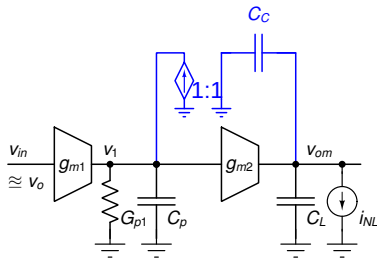


$$H_o(s)_{\text{Indirect Miller}} = \frac{sC_p + G_p}{g_{m1}}$$

$$H_o(s)_{\text{Miller}} = \frac{s(C_C + C_p) + G_{p1}}{g_{m1}}$$

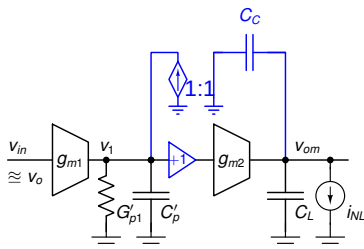
Bias first stage in weak inversion to increase g_{m1}/G_{p1}

Indirect Miller Compensation



$$H_o(s)_{\text{Indirect Miller}} = \frac{sC_p + G_p}{g_{m1}}$$

Buffer between the two stages

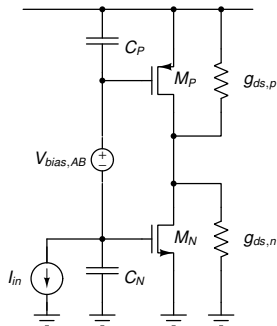
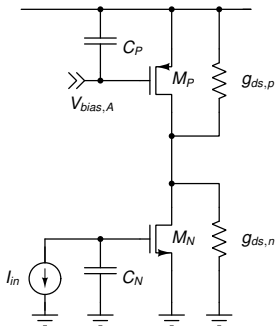


$$H_o(s)_{\text{Indirect Miller, Buffer}} = \frac{sC'_p + G'_p}{g_{m1}}$$

$$C'_p < C_p, G'_p < G_p$$

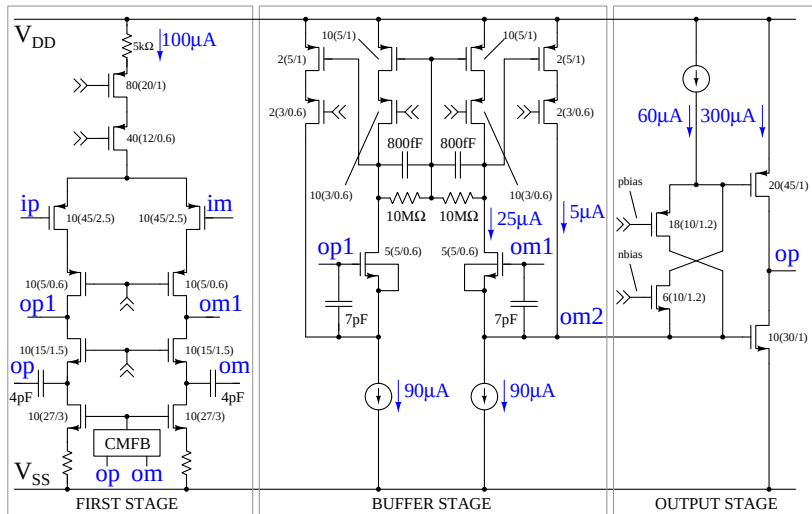
Reduce inherent nonlinearity of the output stage

- Large quiescent current
- High g_m/g_{ds} of second stage:
 - Non-minimum channel length
 - Low overdrive
 - Gascodes: not possible for rail-to-rail
- Use a push-pull stage instead of conventional common-source stage:

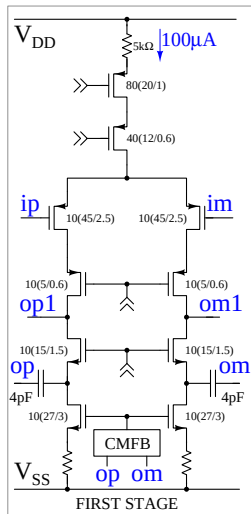


Transistor-level details of the opamp

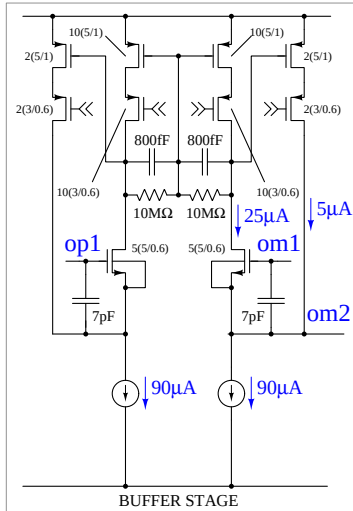
Opamp design for reduced nonlinearity



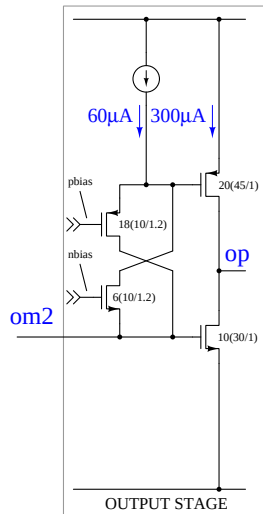
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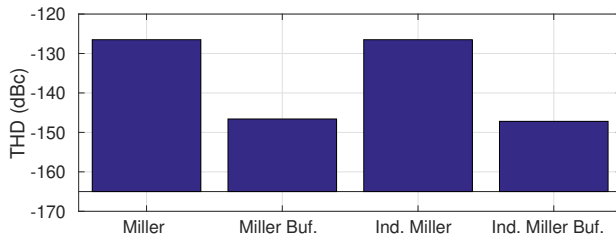
Opamp design for reduced nonlinearity



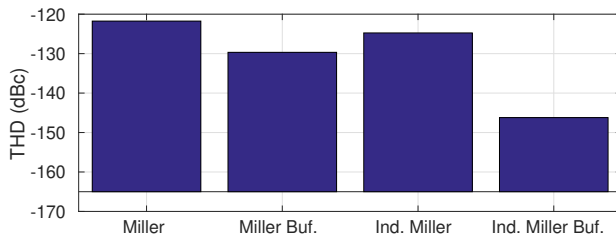
Simulation and measurement results

Simulation results: Comparison of opamp architectures

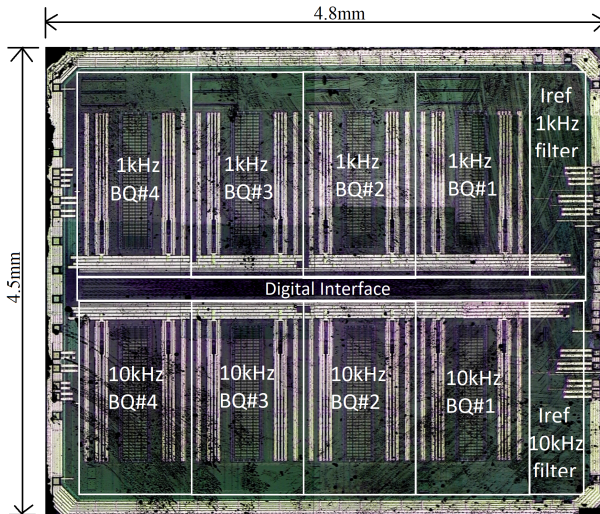
THD: $10 V_{ppd}$ at 1 kHz



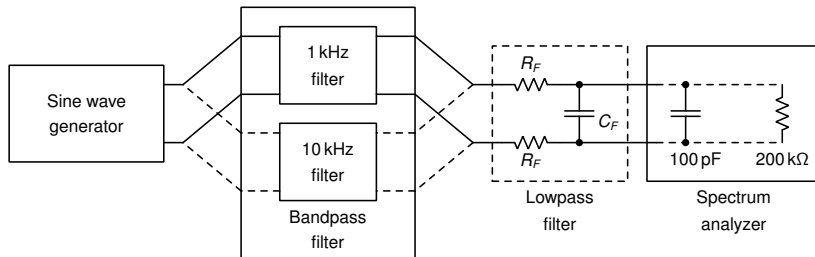
THD: $10 V_{ppd}$ at 10 kHz



Chip photograph

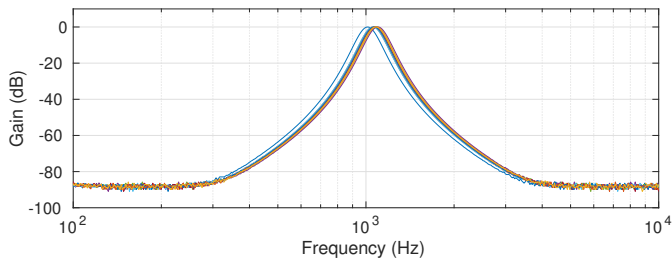


Measurement setup

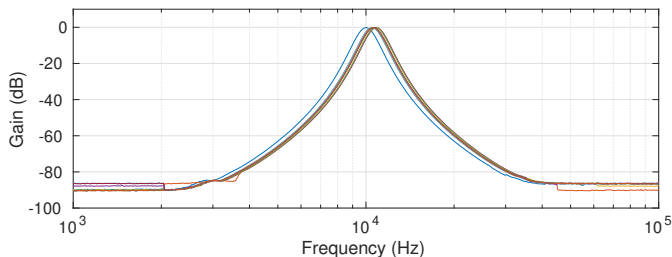


Measured frequency response of 10 devices

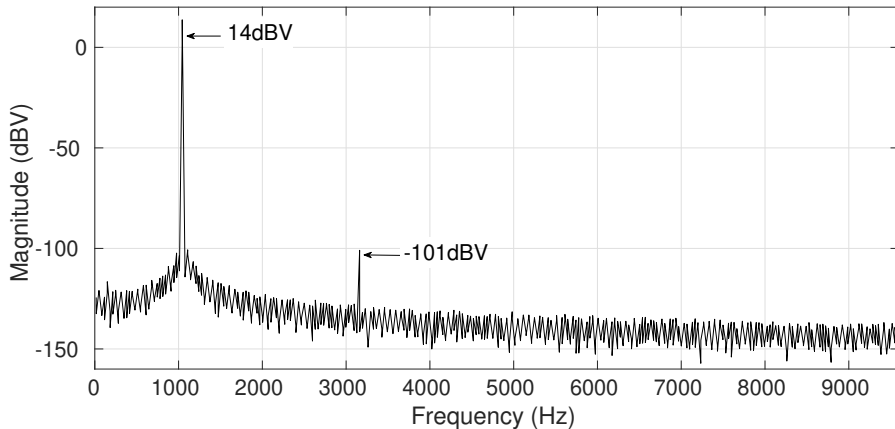
- 1 kHz:



- 10 kHz:

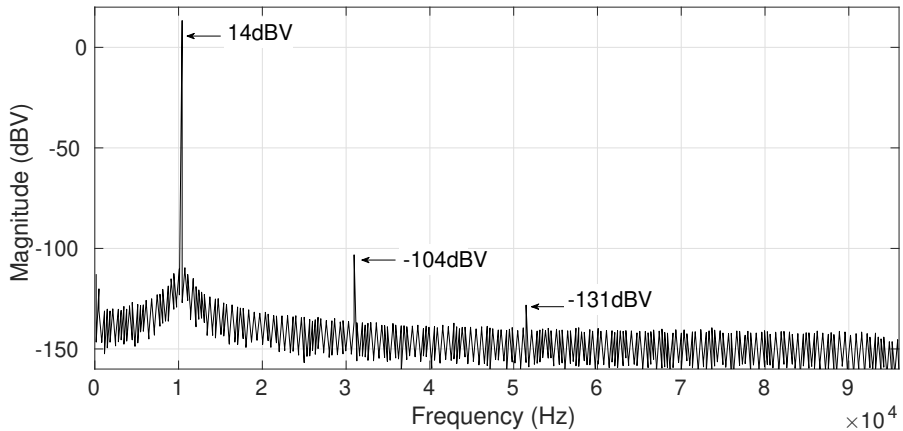


Measured THD: $10 V_{ppd}$ at 1 kHz



THD = -115 dBc

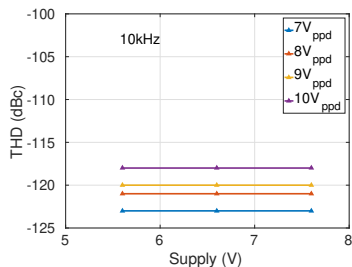
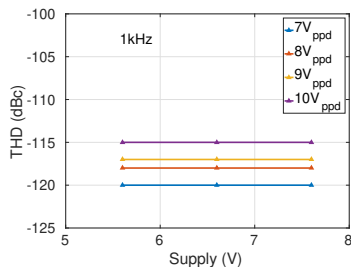
Measured THD: $10 V_{ppd}$ at 10 kHz



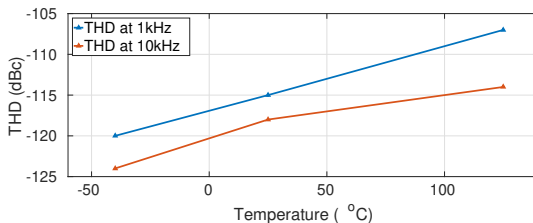
THD = -118 dBc

Measured THD across supply voltage and temperature

- THD vs supply and signal amplitude at 25°C



- THD vs temperature, 10 V_{ppd} signal, 5.6 V supply.



Reference	Tech. (μm)	Order	Freq. (kHz)	Attn. $2f_0$ (dB)	Swing (V_{pp})	Supply (V)	Power (mW)	THD (dBc)	Noise (μV_{rms})	Area mm^2
This work	0.6	8	1	60	10 (Diff)	5.6	65	-115	44.5	11
	0.6	8	10	60	10 (Diff)	5.6	65	-118	44	11
JSSCC '92	1.6	2	10	3	3.66	5	79	-91	31.3	1.4
JSSCC '93	2	5	2	0	4	5	40	-90	100	7
JSSCC '07	0.18	2	2	0	0.6 (Diff)	0.8	1.2	-90	134	0.7
ASIC '98	0.5	5	100	0	2 (Diff)	3.3	43	-75	158	1.53
ICCDSC '06	0.35	4	10	6	2	2.5	48	-65	NA	0.11
MWSCAS '02	0.5	3	4.9	4	2	5	NA	-70	NA	1.5

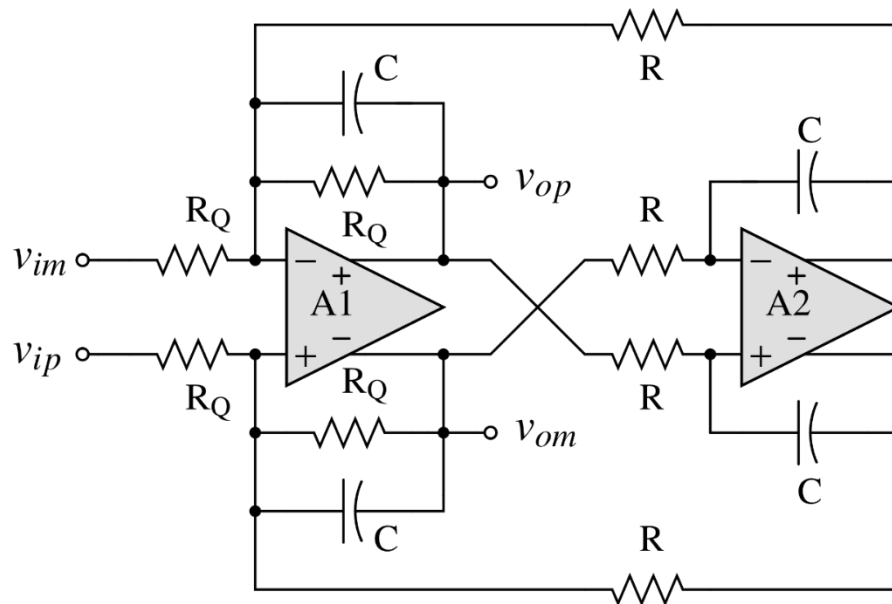
- High attenuation achieved at $2f_0$ with high linearity
- The source obtained can be used to measure 18 - 20 bit ADCs
- Integrated noise over 96 kHz : $44.5 \mu\text{V}_{\text{rms}}$ (1 kHz) and $44 \mu\text{V}_{\text{rms}}$ (10 kHz)

Getting to -140 dB THD

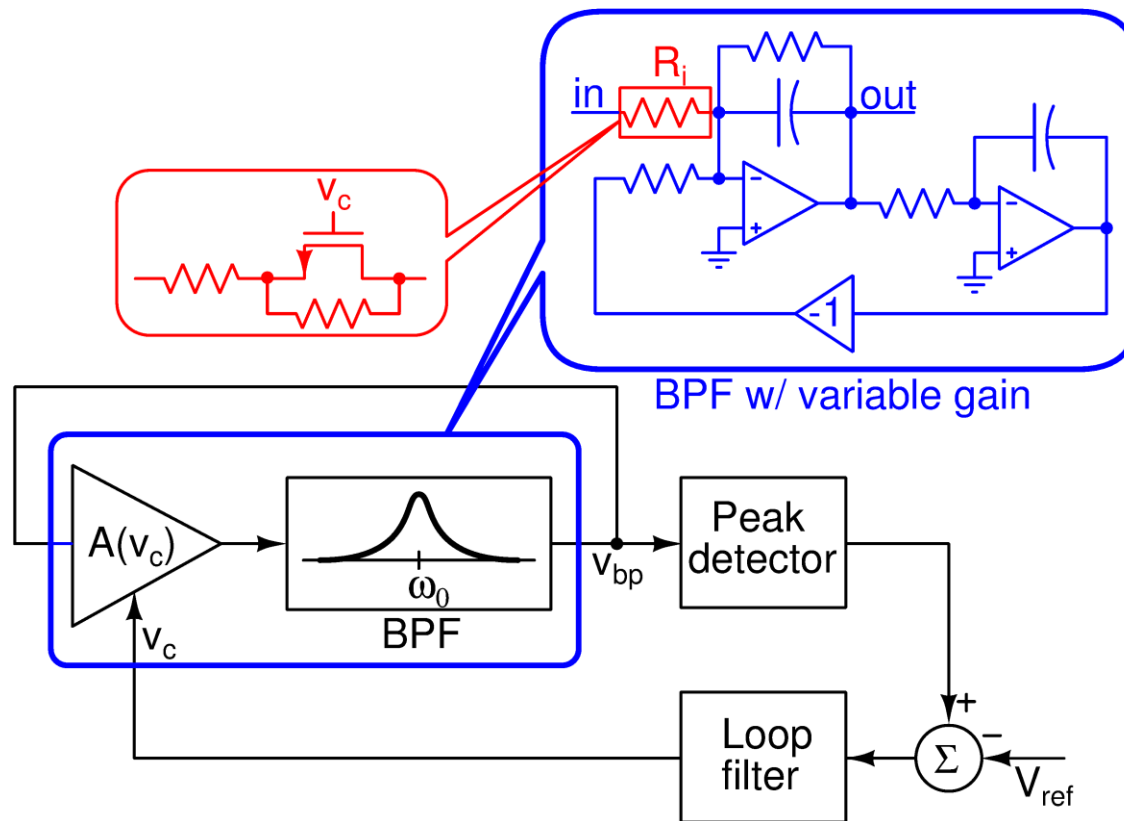
- Problem statement
- Low-distortion band-pass filter
- Low-distortion oscillator
- Distortion mitigation techniques
 - Capacitor nonlinearity cancellation
 - Regulated cascode output stage for the opamps
 - ESD diode nonlinearity
- Results

- IC solution to test linearity of > 18 bit ADCs
 - THD < -140 dBc (1kHz/10kHz) for 10Vppd output at 5.6V supply
 - Small form factor and test scalability
- Two approaches:
 - DAC to generate a sinusoid (THD ~ -80 dBc)
+ active band-pass filter(HD2 attenuation ~ 60 dB)
 - Sinusoidal oscillator(band-pass filter in a positive feedback loop)

- 8th order active-RC BPF
- 4 stages, each with $Q=4$
- Two-stage opamp
 - Buffer after the first stage to reduce nonlinear current of the parasitic cap of second stage
- THD $\sim$ -115/-118 dBc at 1/10 kHz for 10Vppd output

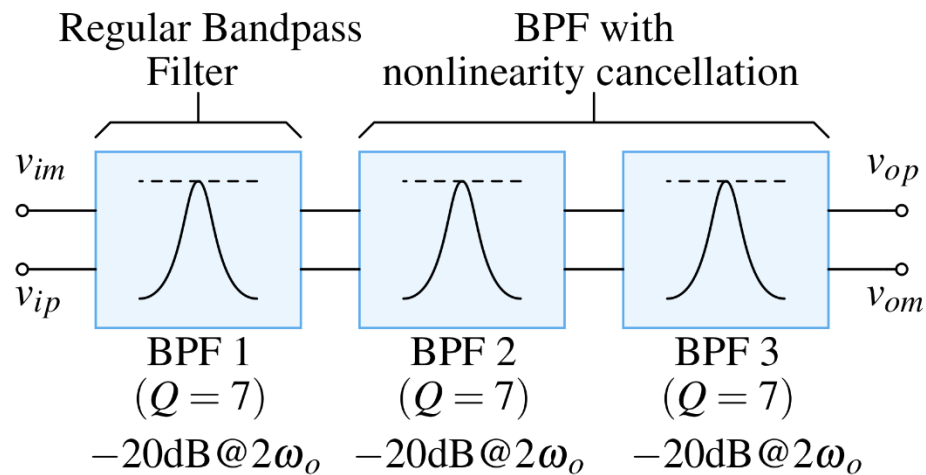


- BPF in a loop with amplitude stabilization
- THD $\sim$ -115/-121 dBc at 1/10 kHz for 8Vppd output

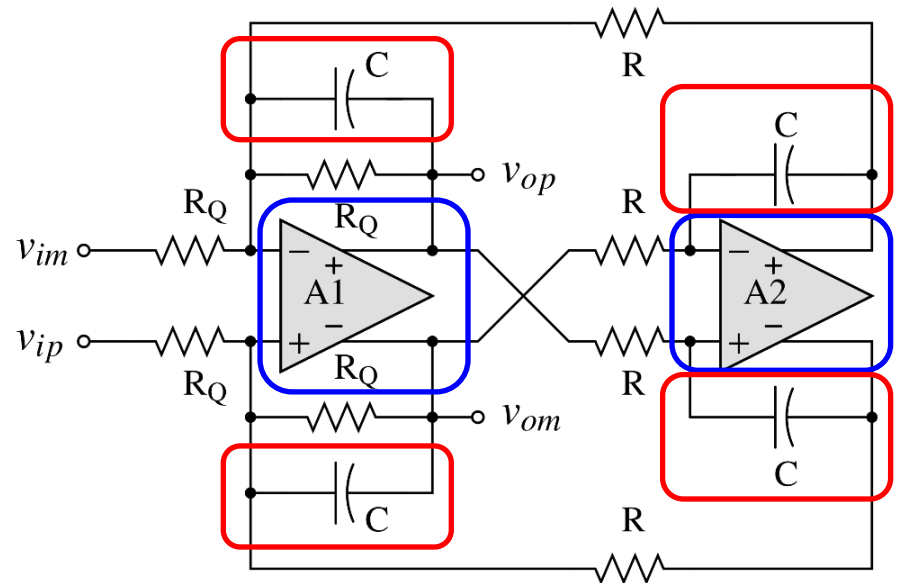


- Capacitor nonlinearity
 - Not suppressed by loop gain
 - Cancelled by injecting opposite nonlinear current
- Opamp output conductance nonlinearity
 - Suppressed using local negative feedback (Regulated cascode)
- ESD diode nonlinearity
 - Four-wire connection

Proposed solution 1: 6th Order(3-stage) BPF

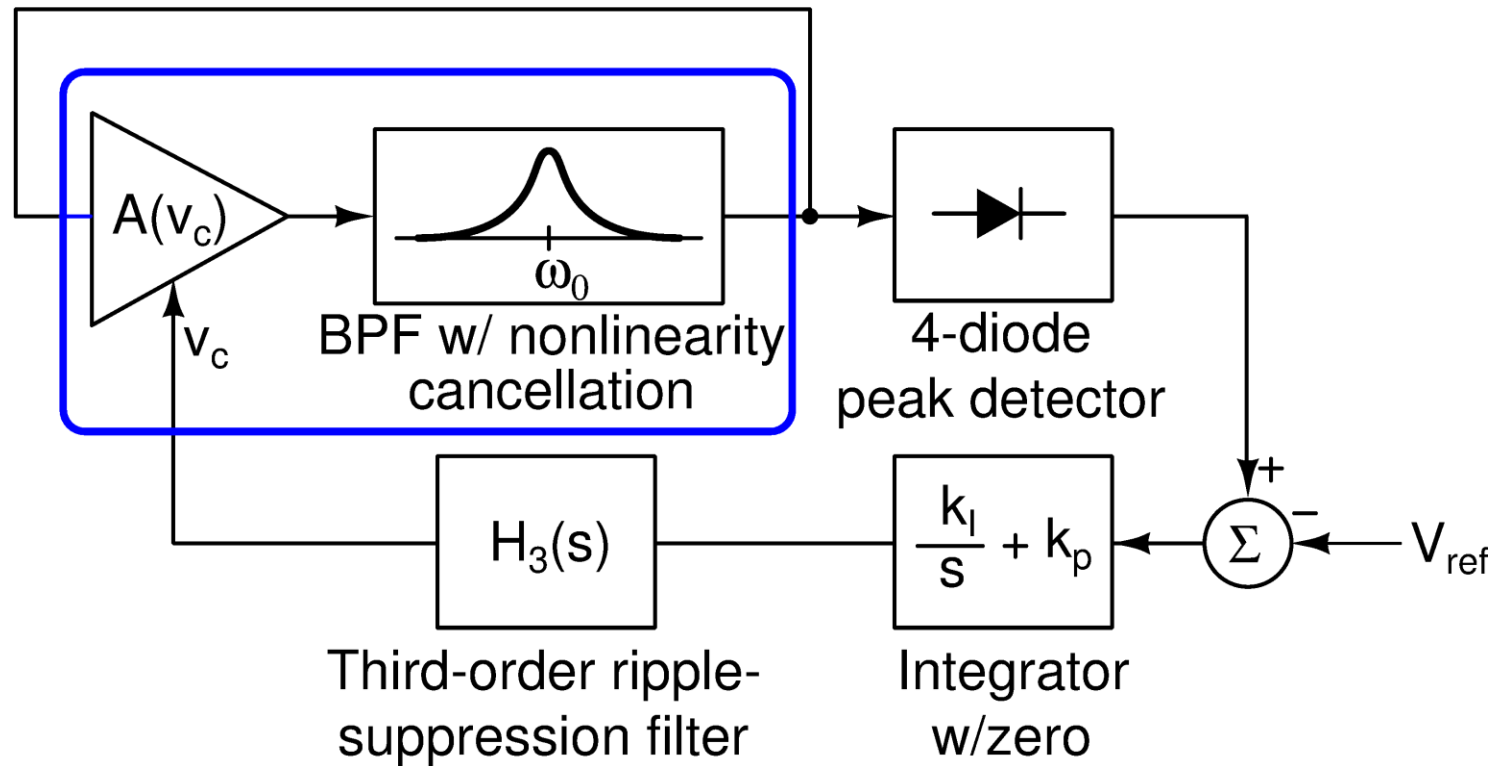


Cancel capacitor nonlinearity

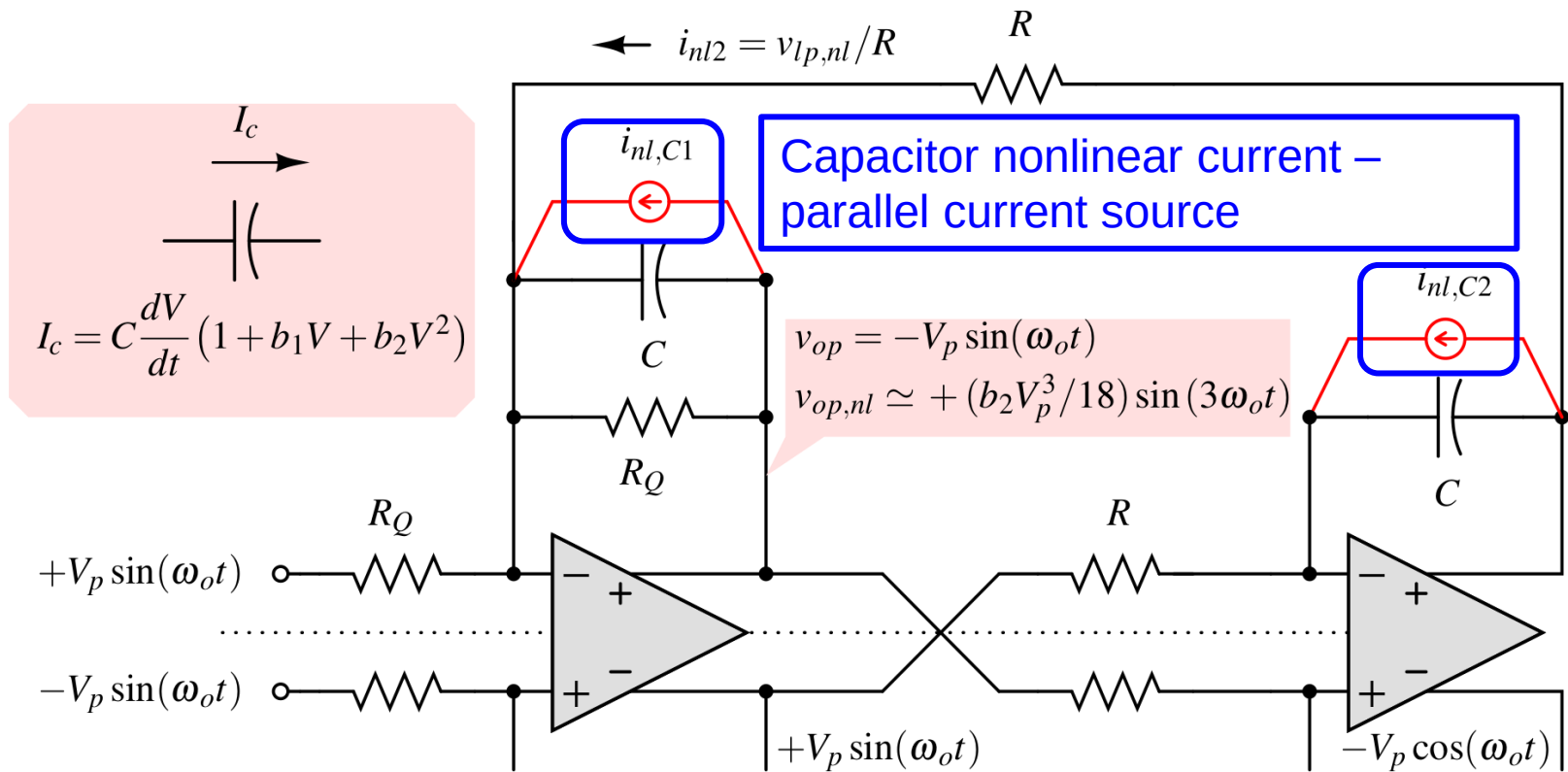


Limit V_{DS} swing (Regulated cascode)

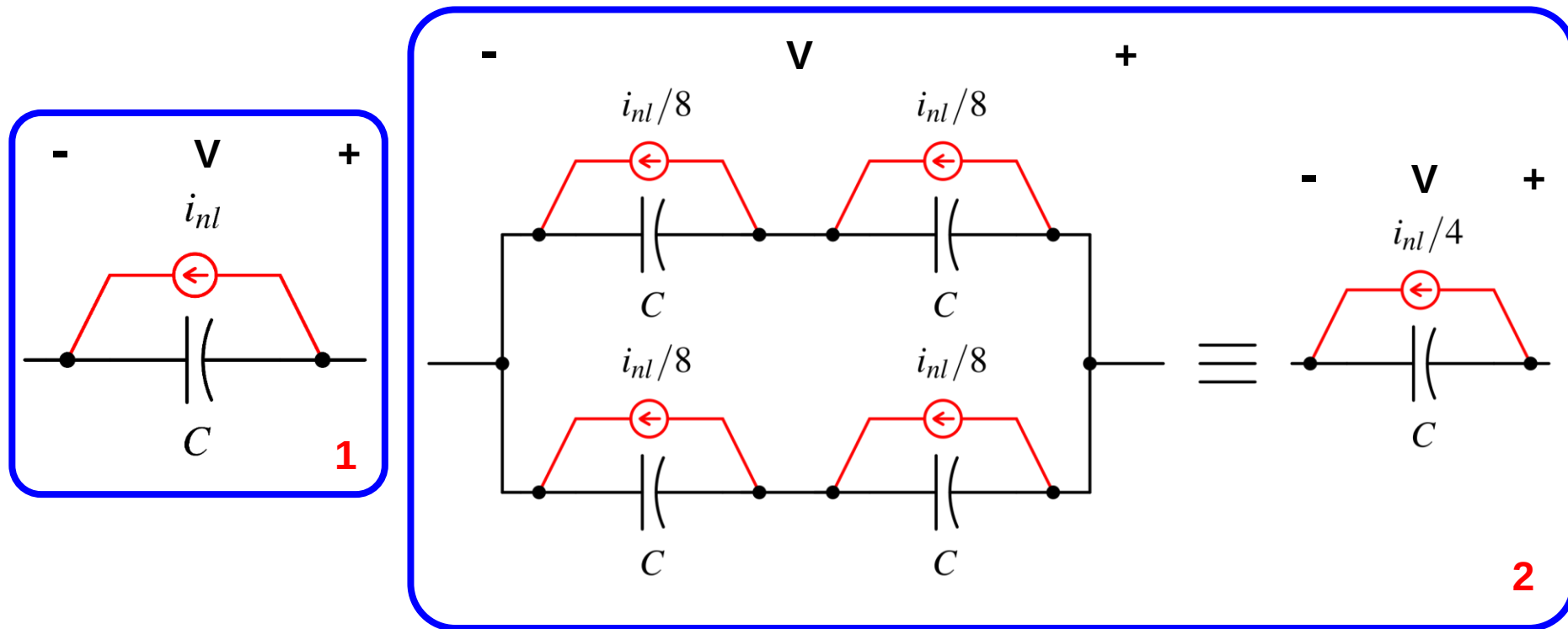
Oscillator (BPF + Amplitude stabilization)



BPF Distortion

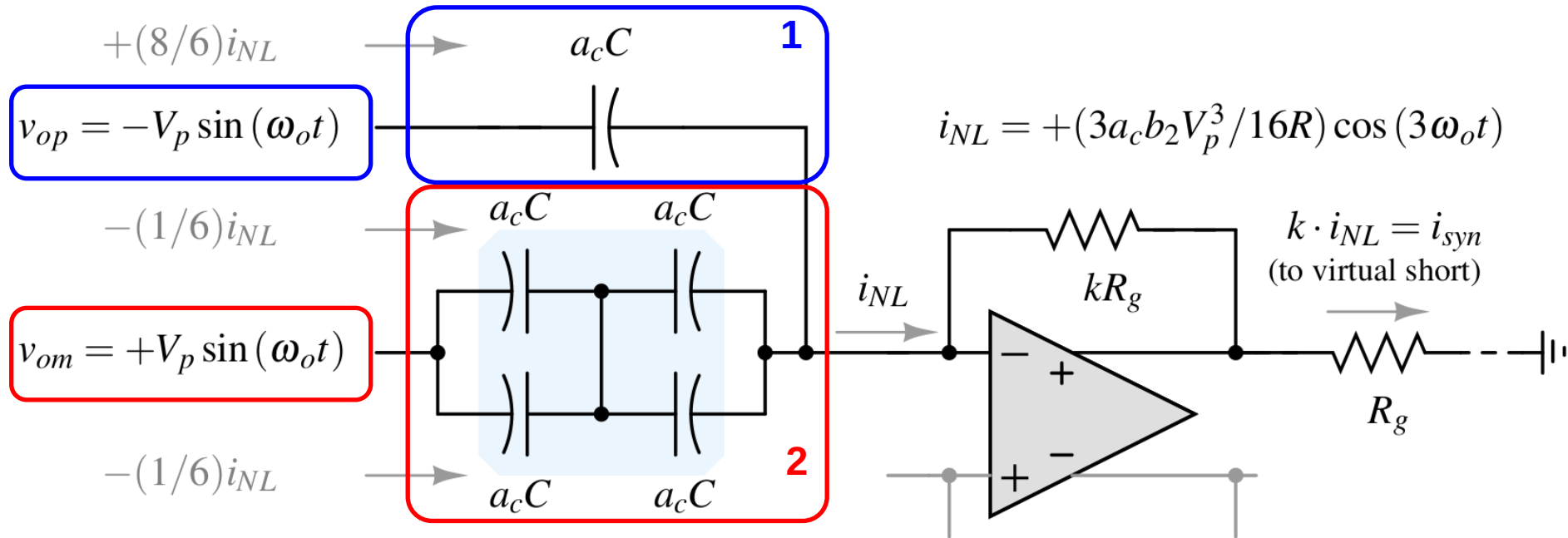


Extracting nonlinear current



Identical linear part; Scaled nonlinear part

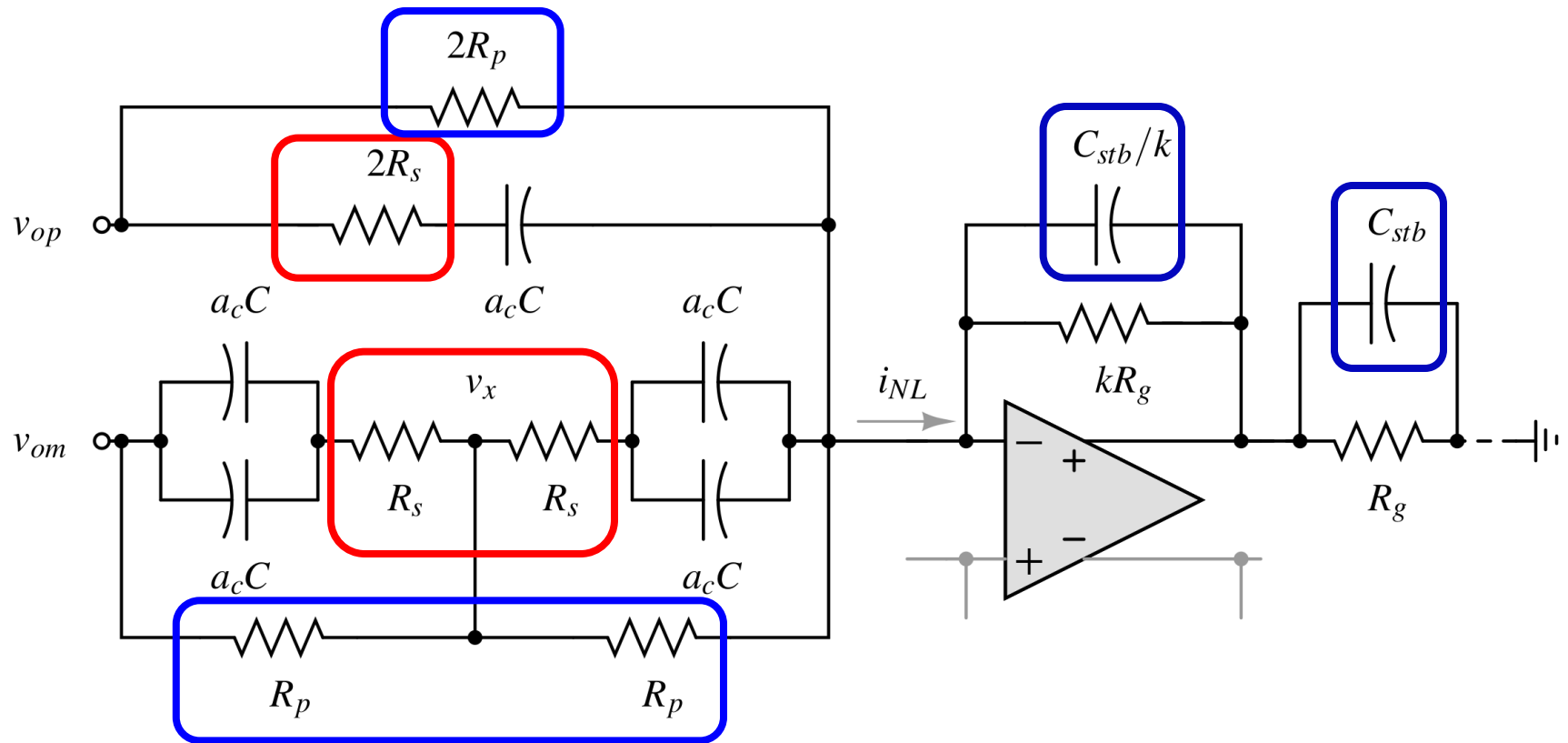
Extracting nonlinear current



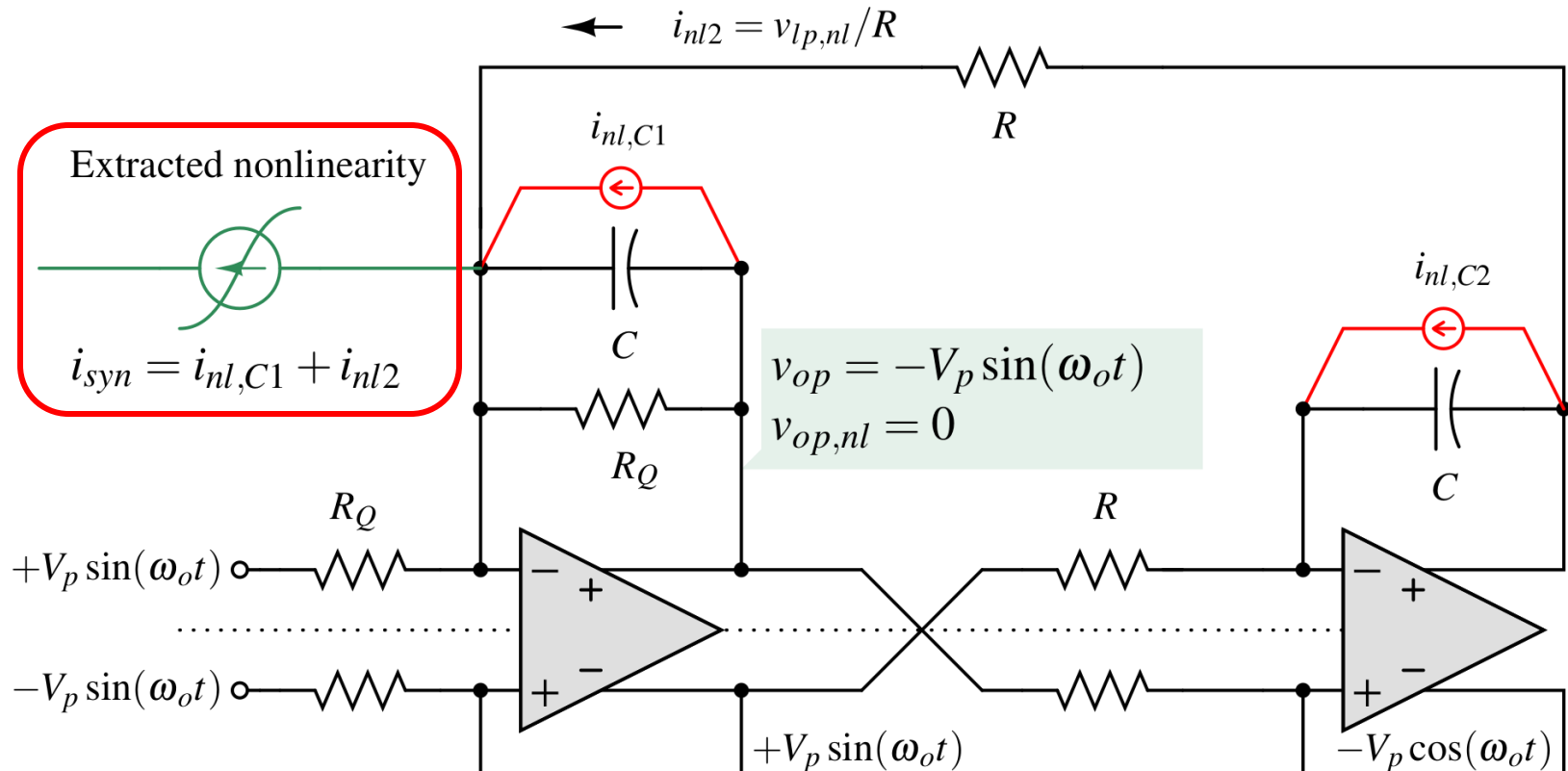
Linear part cancelled; Nonlinear part remains

Smaller $a_c \Rightarrow$ Smaller area; **More excess noise**

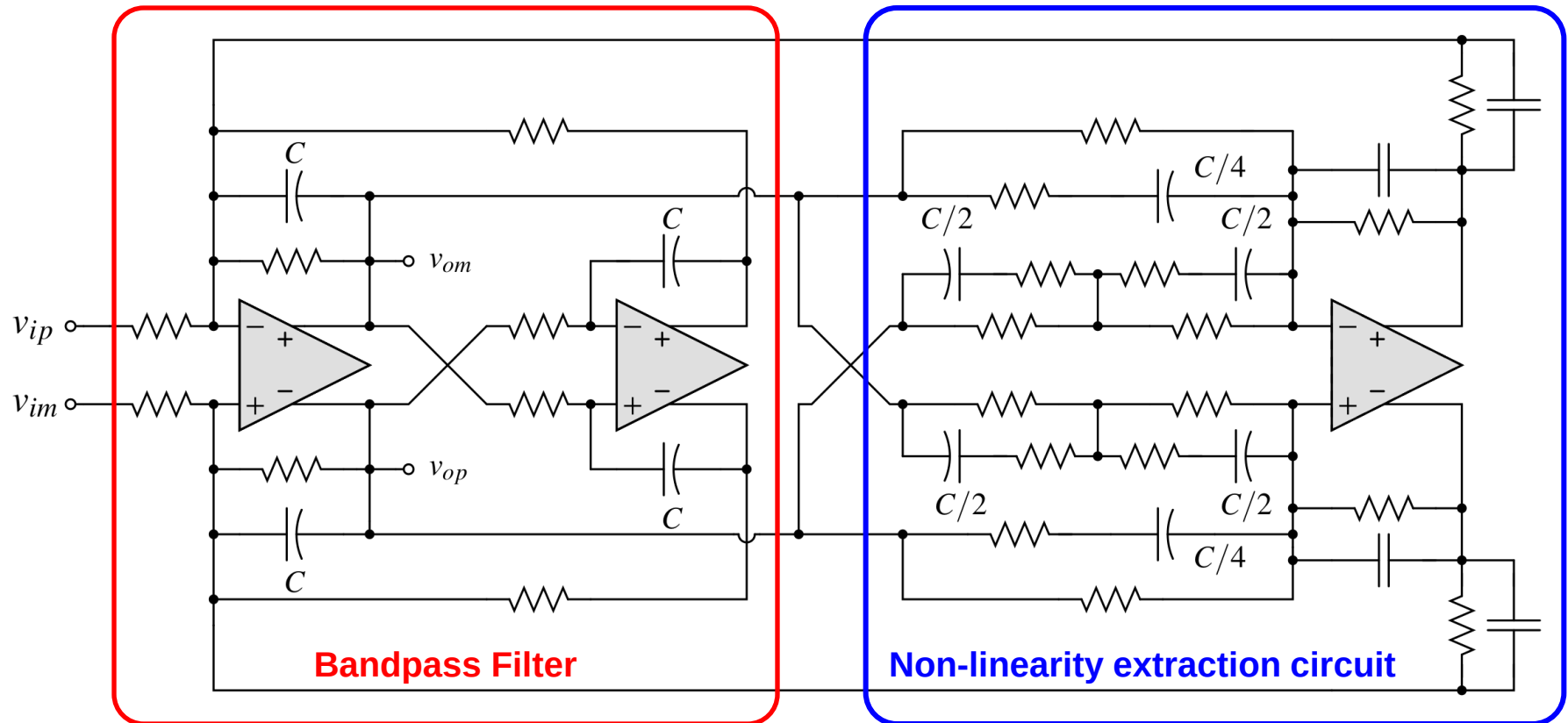
More practical circuit : + C_{stb} , R_g , R_p , R_s



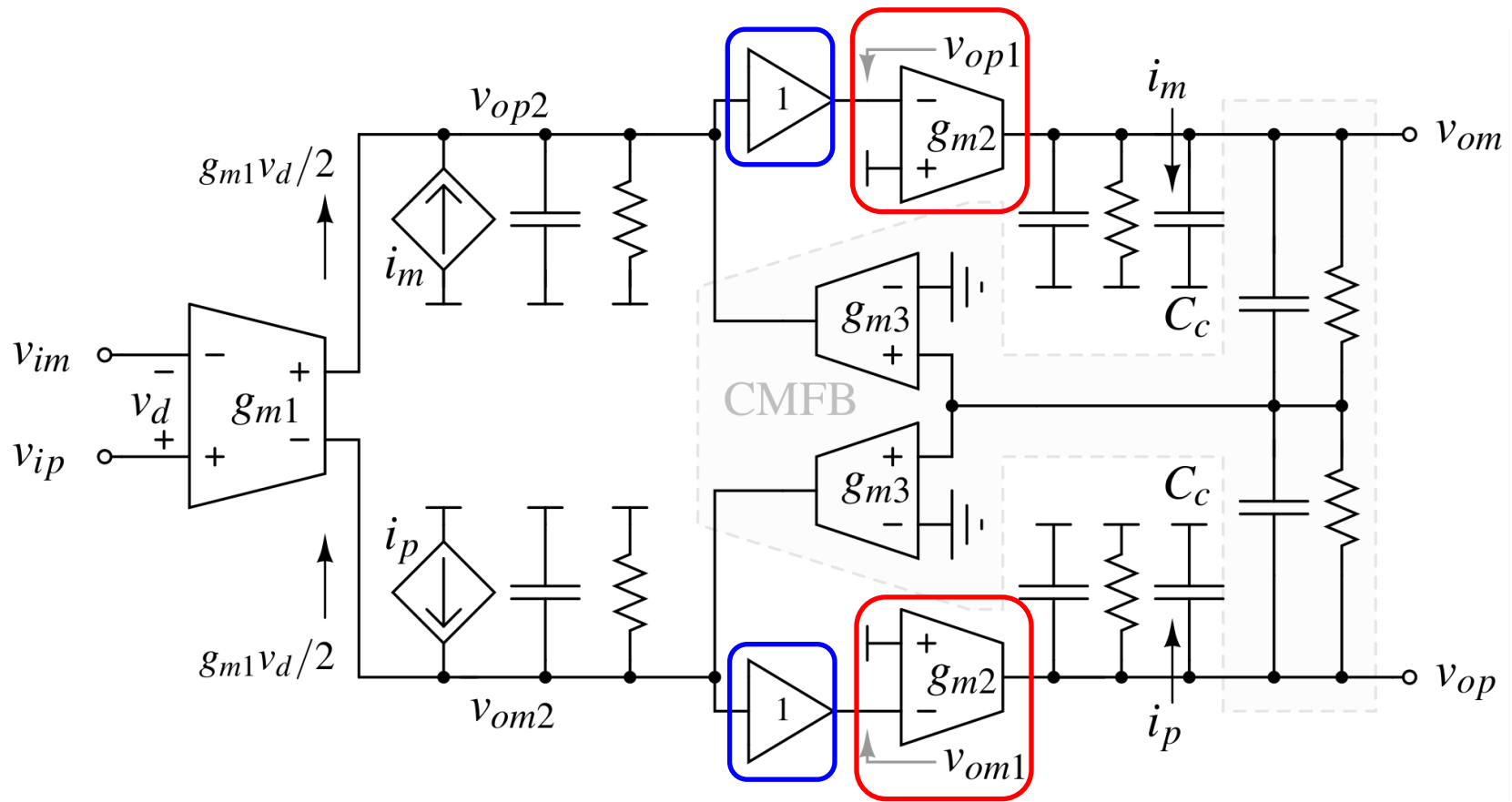
$I_{syn} = i_{nl2} + i_{nlC1}$; Nonlinearity cancelled



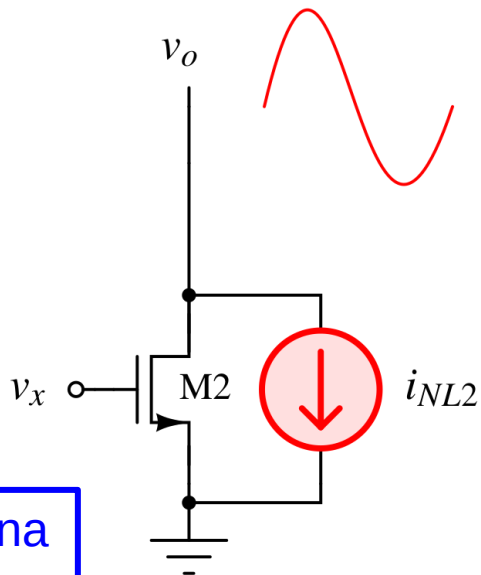
Complete improved single stage BPF



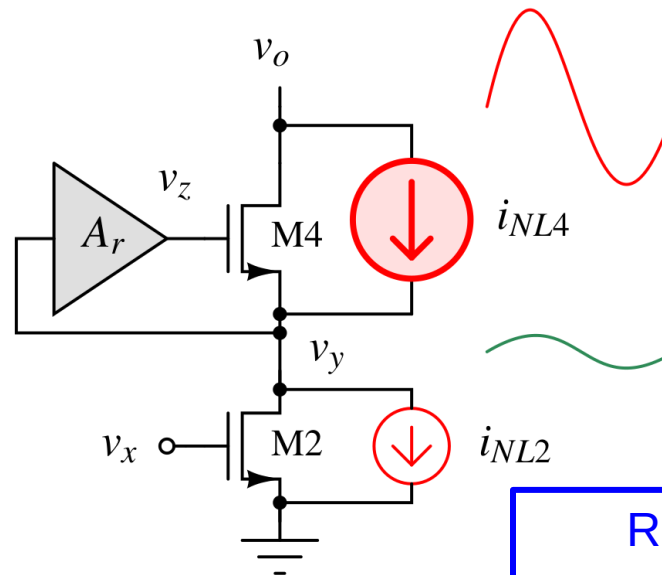
Opamp Architecture



Output conductance nonlinearity mitigation



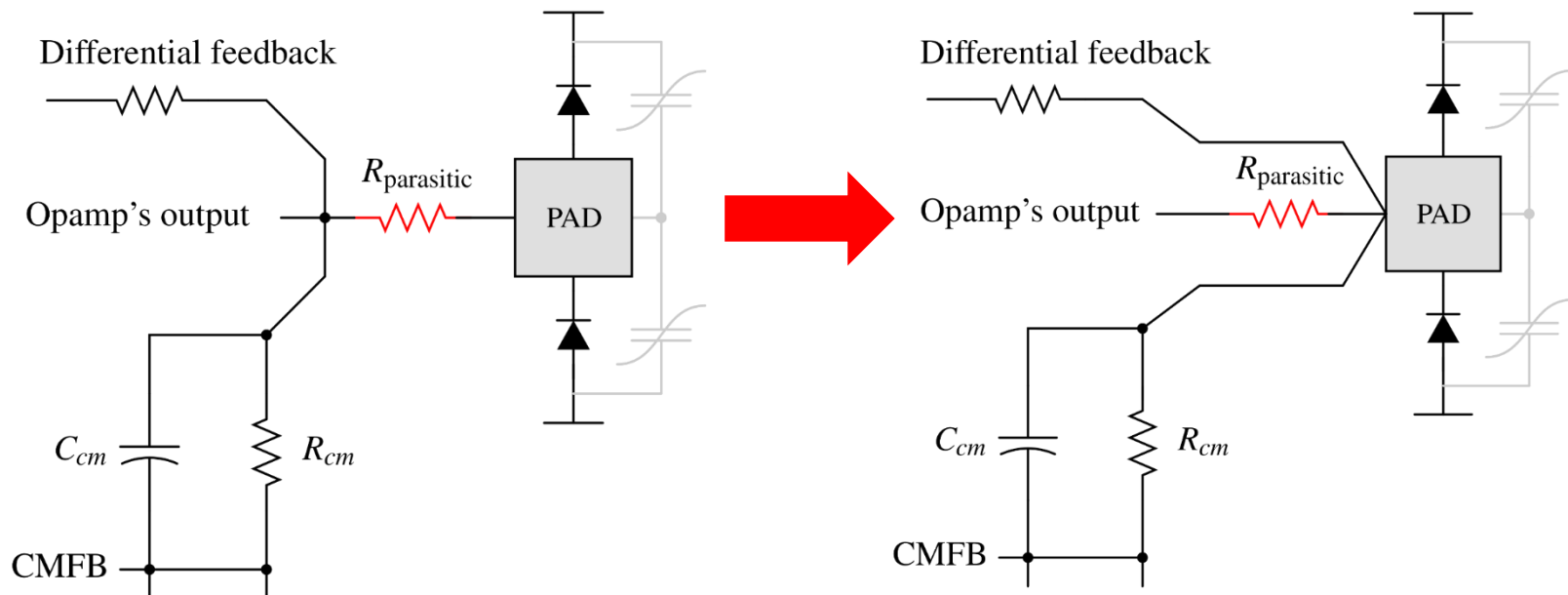
Conventional



Regulated cascode

- Small swing at node $v_y \Rightarrow$ negligible i_{NL2}
- Distortion i_{NL4} is suppressed by factor of $A_r g_{m4(1)} / g_{ds2(1)}$

ESD diode nonlinearity

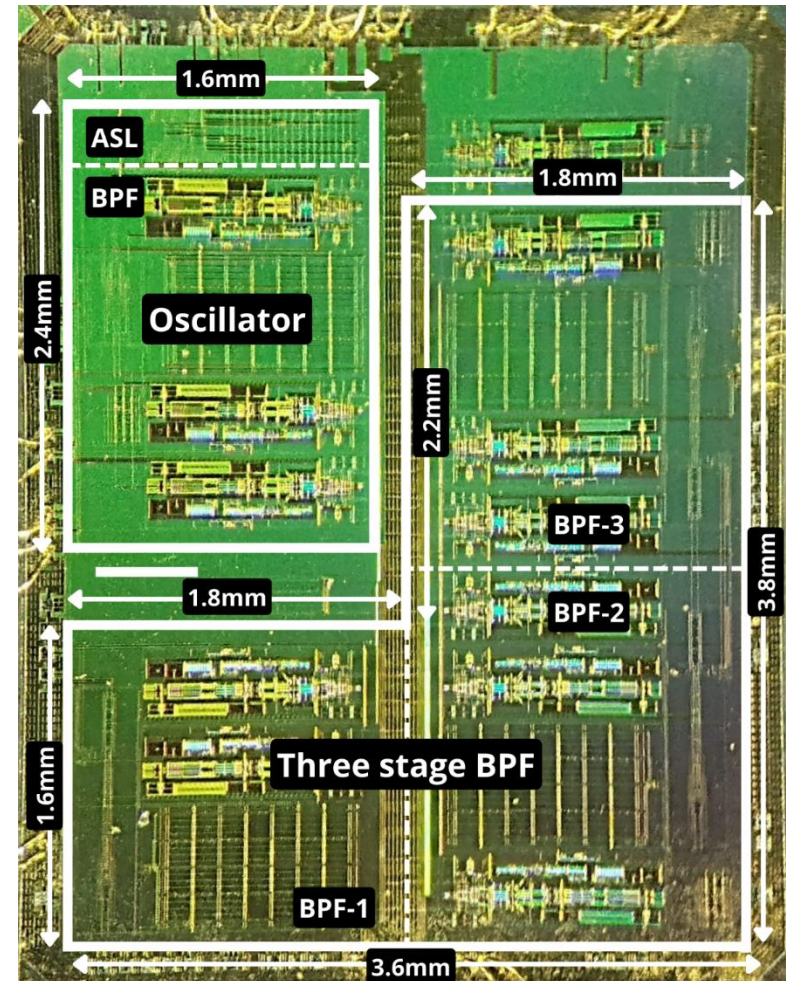


- **Four-wire connection suppresses the effect of $R_{\text{parasitic}}$.**
- **Both HD2 and HD3 nonlinearity due to ESD diode reduced**

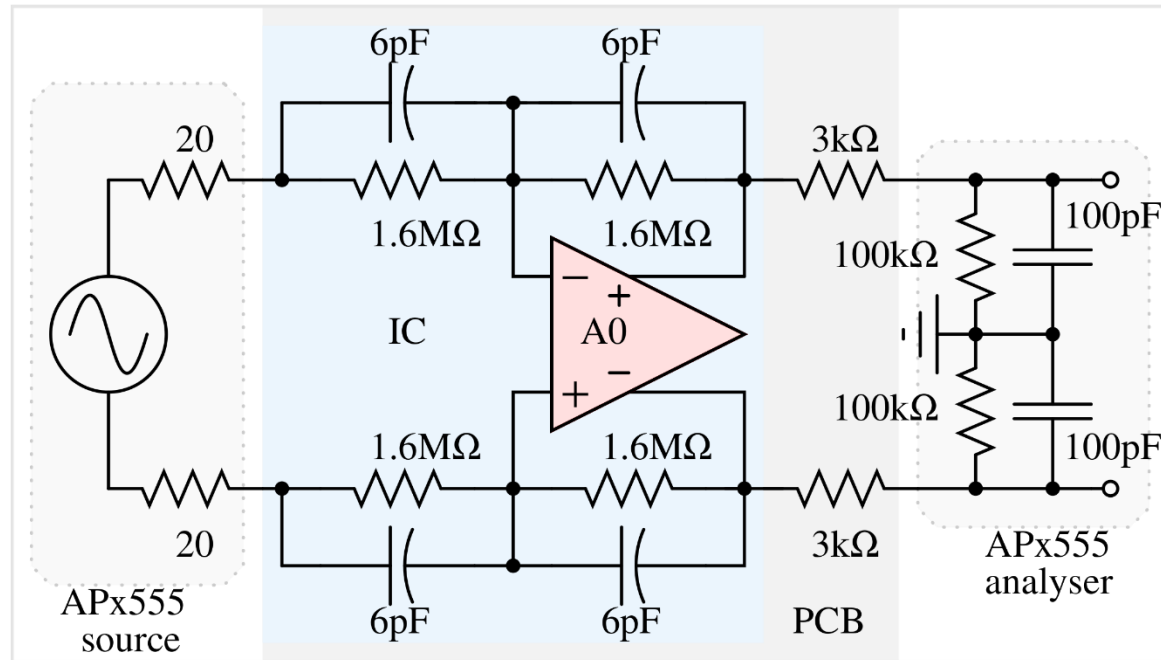
- Sixth-order band-pass filter
- Sinusoidal oscillator
- Switchable to 1kHz/10kHz

+ Test chips

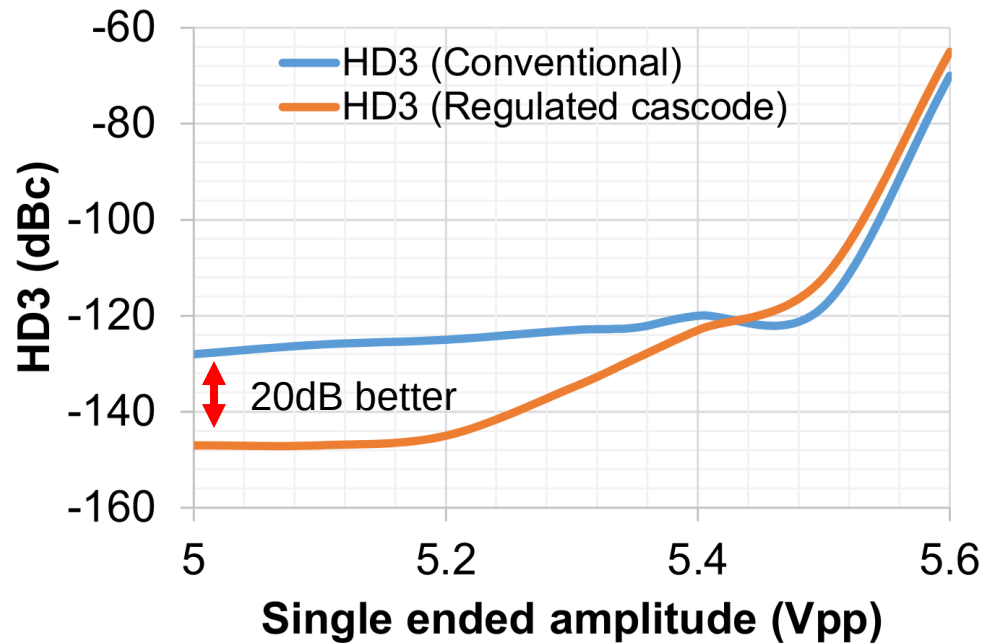
- Amplifier: conventional and regulated-cascode output
- 2nd order filter: With and w/o nonlinearity cancellation



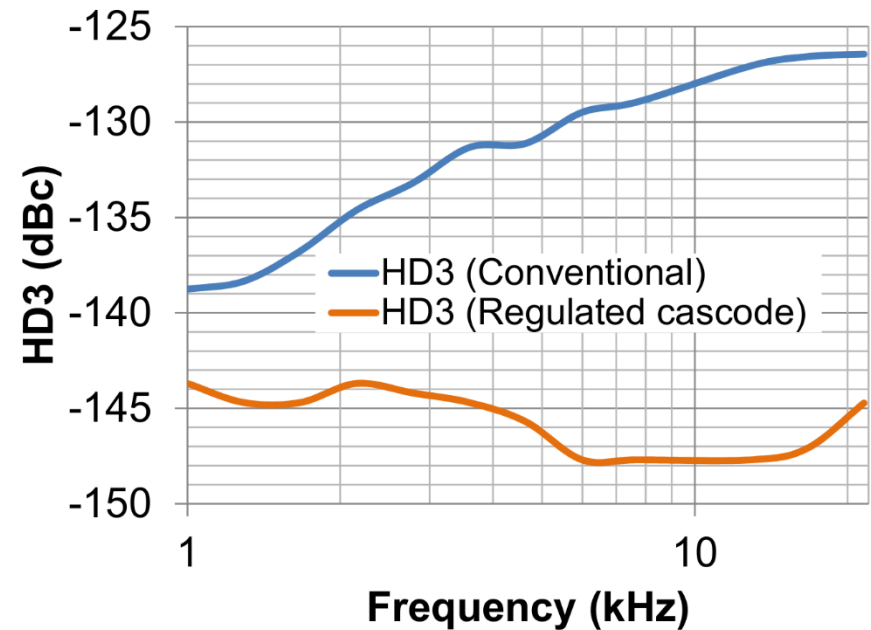
Amplifier test chip



Amplifier test chip: Measured results

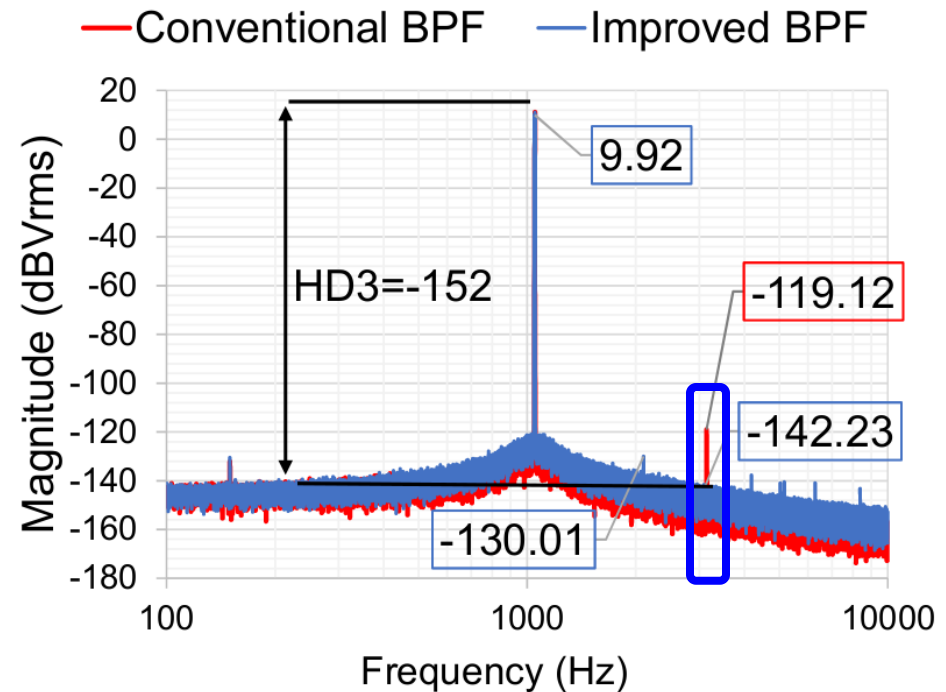
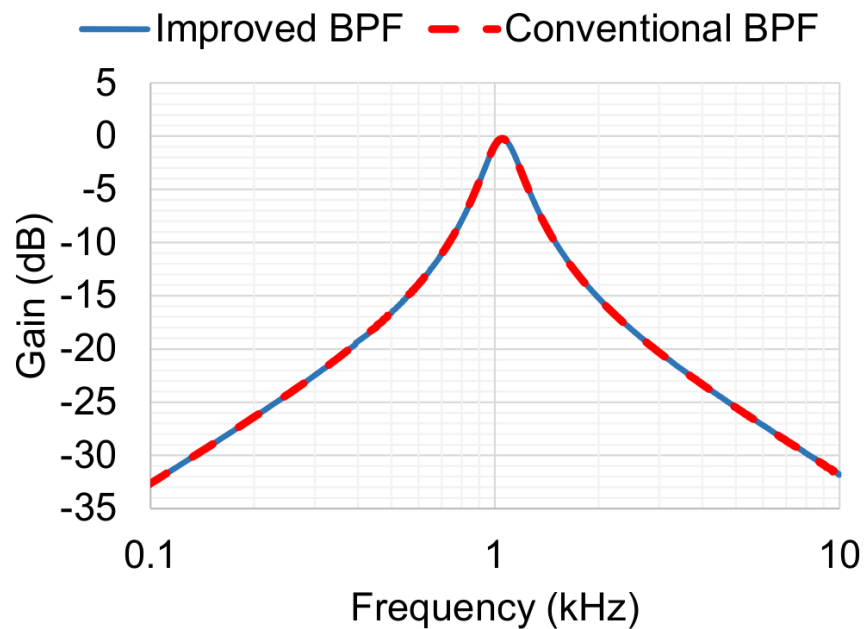


At 10
kHz

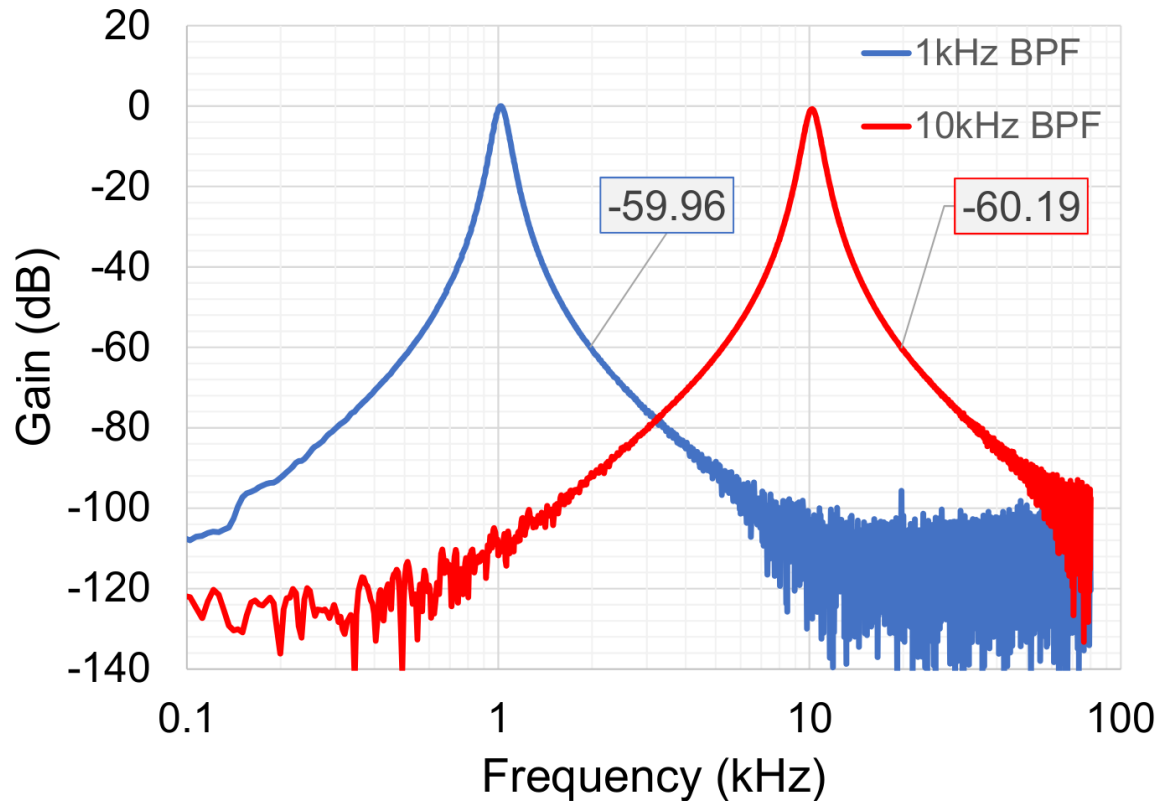


5 Vpp single-ended
amplitude

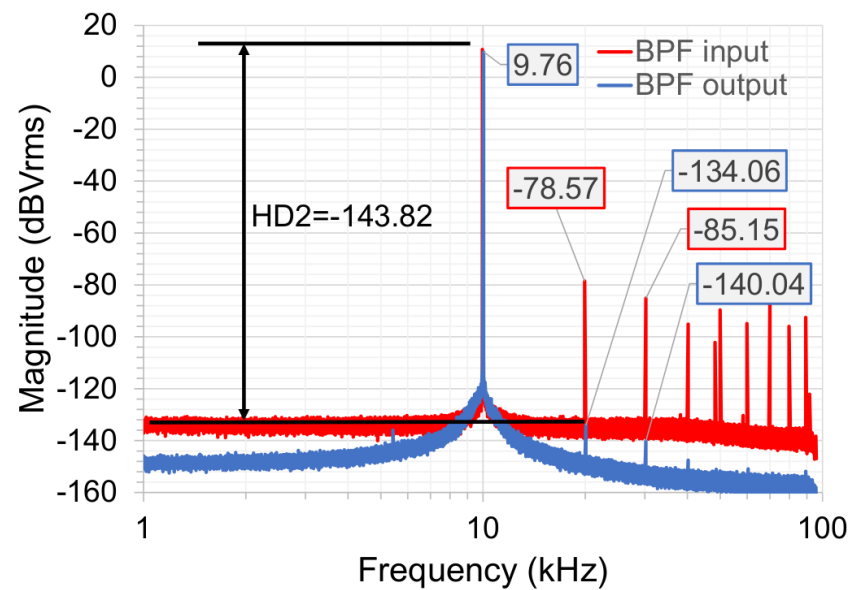
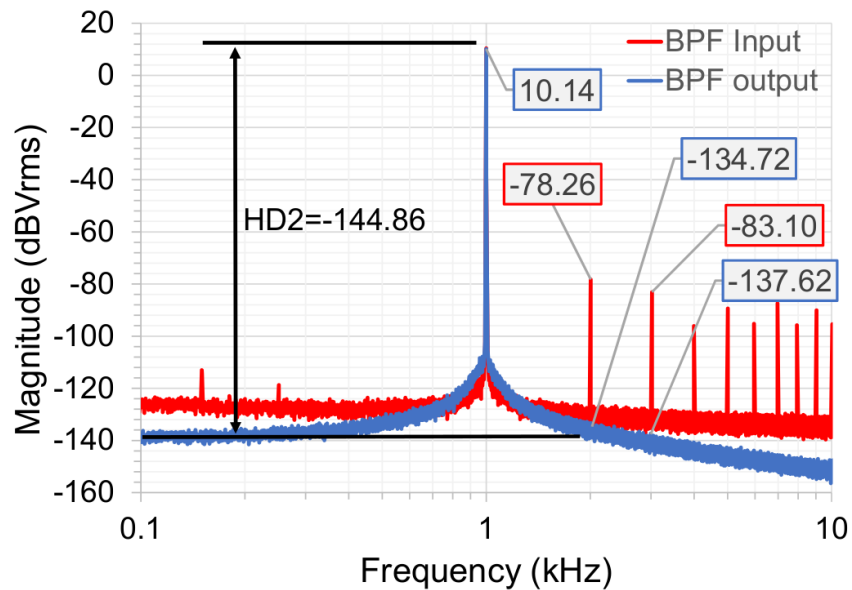
Second-order BPF test chip: Measured results



Sixth-order BPF: Measured magnitude response



Measured THD – 1kHz and 10kHz

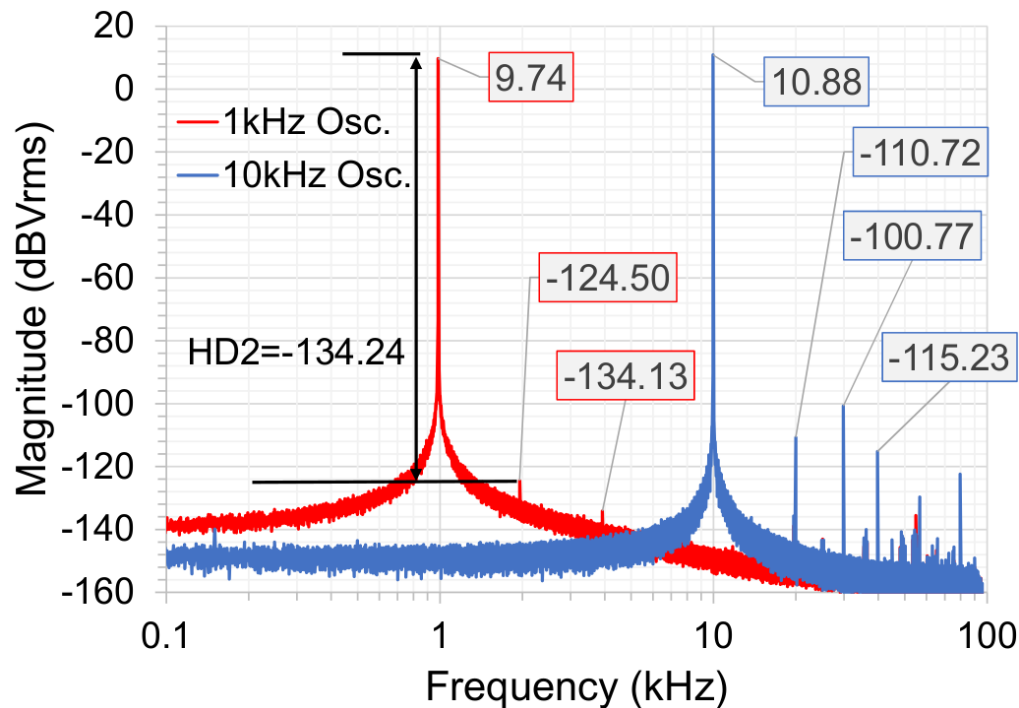


THD < -140 dBc

BPF performance summary

Parameters	This work	[1] 2019 TCAS-I	[3] 1992 JSSC	[4] 1993 JSSC	[5] 2019 ISSCC	[6] 2022 ISSCC
Section	Filter	Filter	Filter	Filter	Dac+Amp	Dac+Amp
Process (um)	0.6	0.6	1.6	2	0.153	0.028
Filter order	6	8	2	5	-	-
Freq (kHz)	1,10	1,10	10	2	1	1
Swing (Vpp)	10(diff)	10(diff)	3.66	4	2.8	2.8
Supply (V)	5.6	5.6	5	5	1.8	1.8
Power (mW)	62.7	65	79	40	-	-
THD (dBc)	-143,-142	-115,-118	-91	-90	-105	-117
Noise (uVrms)	77.8,58.9	44.5,44.0	31.3	100	-	-
Area (mm.sq)	9.72	11	1.4	7	1.27	1.14

With regulated-cascode output stage and capacitor nonlinearity cancellation



THD = -133.8 dB at 1kHz, -111.1 dB at 10kHz

Parameters	This work	[1] TCAS-I 2019	APx555
Process (um)	0.6	0.6	Bench-top
Freq (kHz)	1,10	1,10	1,10
Swing (Vpp)	10(diff)	8(diff)	10(diff)
Supply (V)	5.6	5	-
Power (mW)	34.7	49.5	-
THD (dBc)	-133.8,-111.1	-114.8,-116.5	-150,-145
Area (mm.sq)	3.84	8.5	-

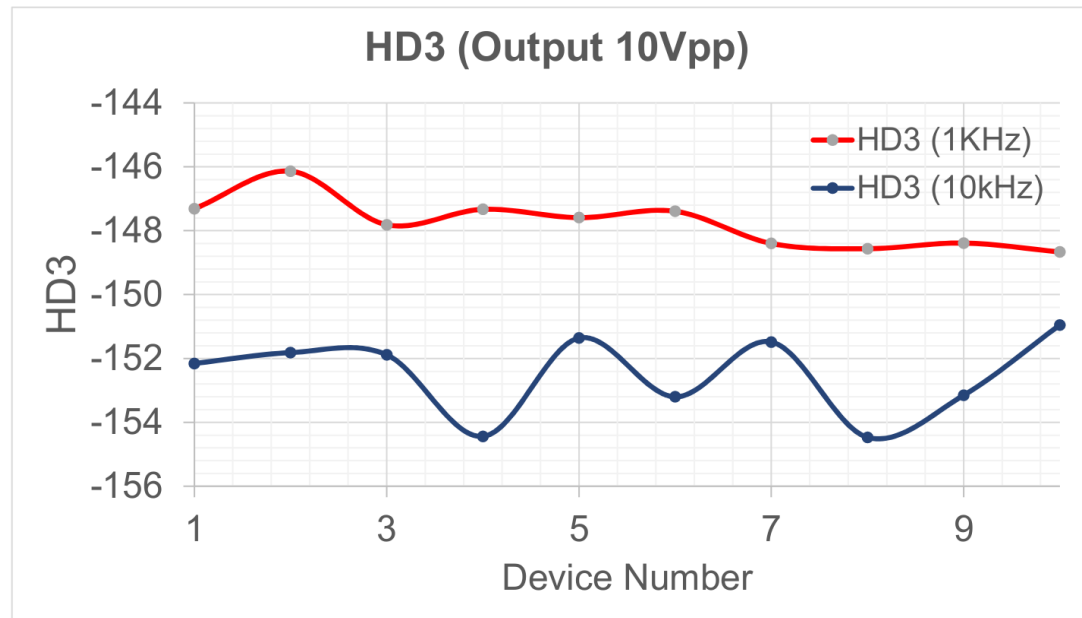
- Two techniques to suppress distortion to $< -140\text{dB}$
 - Capacitor HD3 cancellation technique
 - Regulated cascode output stage for the opamp
- Low-distortion sinusoid generation demonstrated
 - Band-pass filter with -142dBc THD
 - Oscillator with -133dBc THD

References

- [1] S. Kumar et al., “Design Considerations for Low-Distortion Filter and Oscillator ICs for Testing High-Resolution ADCs,” *IEEE TCAS-I*, vol. 66, no. 9, pp. 3393-3401, Sept. 2019.
- [2] E. Sackinger and W. Guggenbuhl, “A High-Swing, High-Impedance MOS Cascode Circuit,” *IEEE JSSC*, vol. 25, no. 1, pp. 289-298, Feb. 1990.
- [3] A. M. Durham et al., “High-Linearity Continuous-Time Filter in 5-V VLSI CMOS,” *IEEE JSSC*, vol. 27, no. 9, pp. 1270-1276, Sept. 1992.
- [4] Un-Ku Moon and Bang-Sup Song, “Design of a Low-Distortion 22-kHz Fifth-Order Bessel Filter,” *IEEE JSSC*, vol. 28, no. 12, pp. 1254-1264, Dec. 1993.
- [5] S. Wen et al., “A -105dBc THD+N (-114dBc HD2) at 2.8VPP Swing and 120dB DR Audio Decoder with Sample-and-Hold Noise Filtering and Poly Resistor Linearization Schemes,” *ISSCC*, pp. 294-295, Feb. 2019.
- [6] Wen, S.H. et al., “A -117dBc THD (-132dBc HD3) and 126dB DR Audio Decoder with Code-Change-Insensitive RT-DEM Algorithm and Circuit Technique for Relaxing Velocity Saturation Effect of Poly Resistors,” *ISSCC*, Vol. 65, pp. 482-483, Feb. 2022.

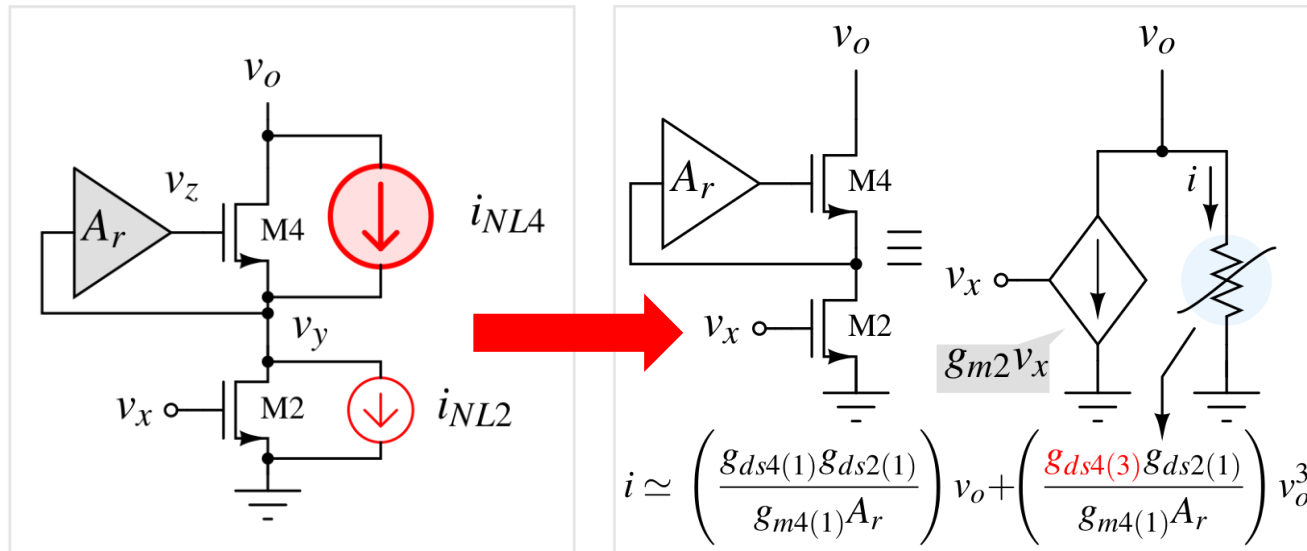
Backup slides

Measurement across multiple devices



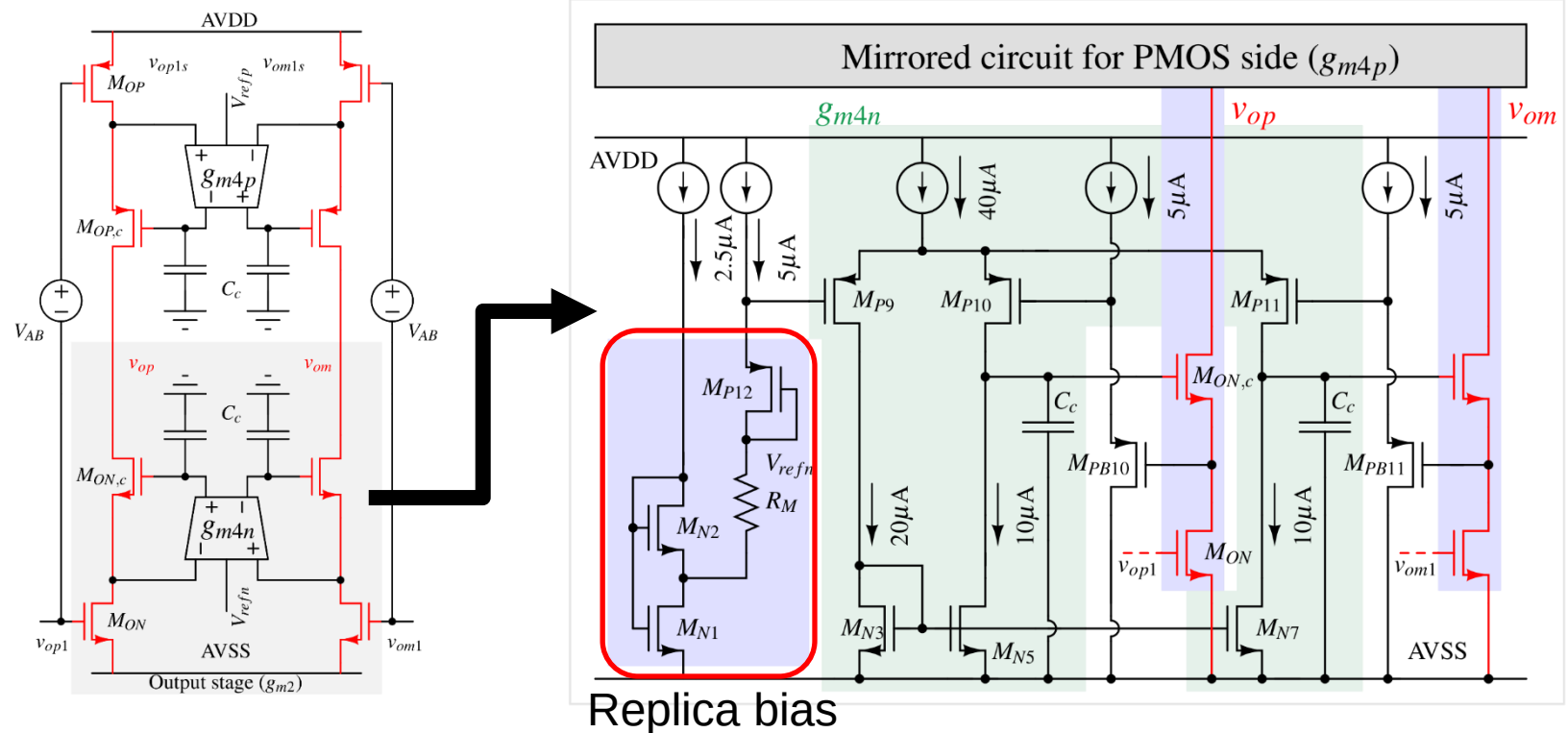
- Low distortion sine wave input from APx555
- 1kHz's third harmonic was lying below noise floor
- The noise in 1kHz is higher than 10kHz. So, bit worse HD3 for 1kHz

Gain boosting improving linearity



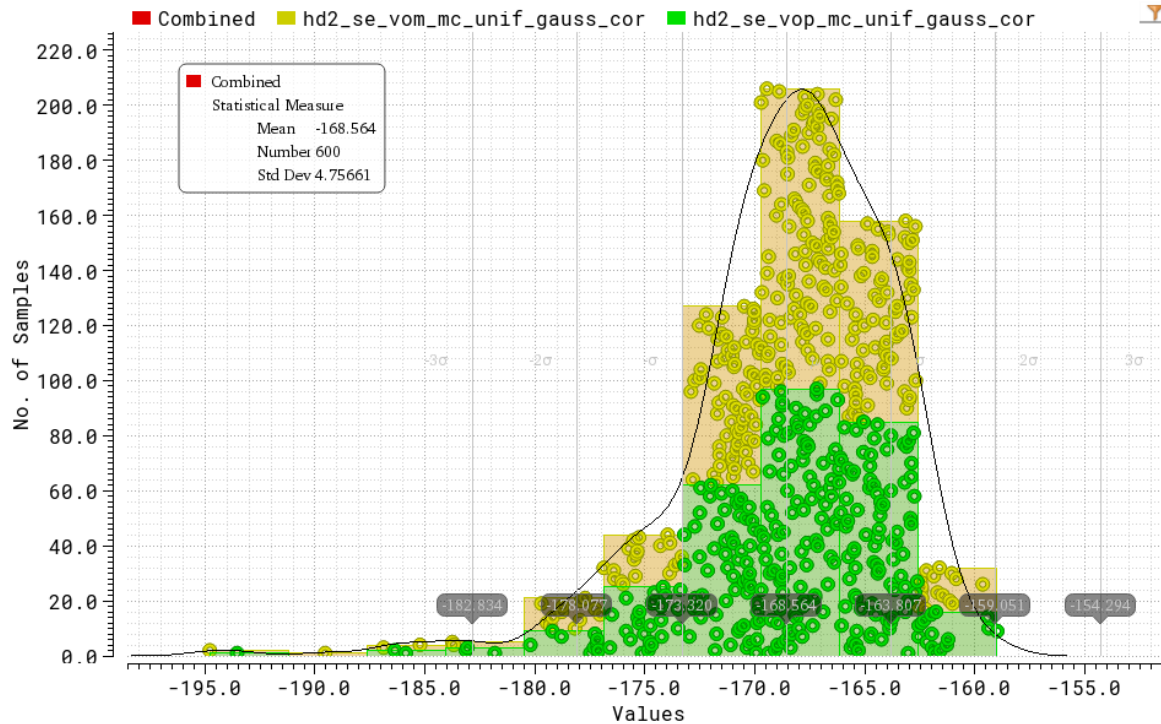
- Very small swing at node $v_y \Rightarrow$ negligible i_{NL2}
- Large i_{NL4} , however A_r pre-distorts v_z ; i_{NL4} goes through $M4$
- Distortion is suppressed by factor of $A_r g_{m4(1)} / g_{ds2(1)}$

Output stage transistor level schematic



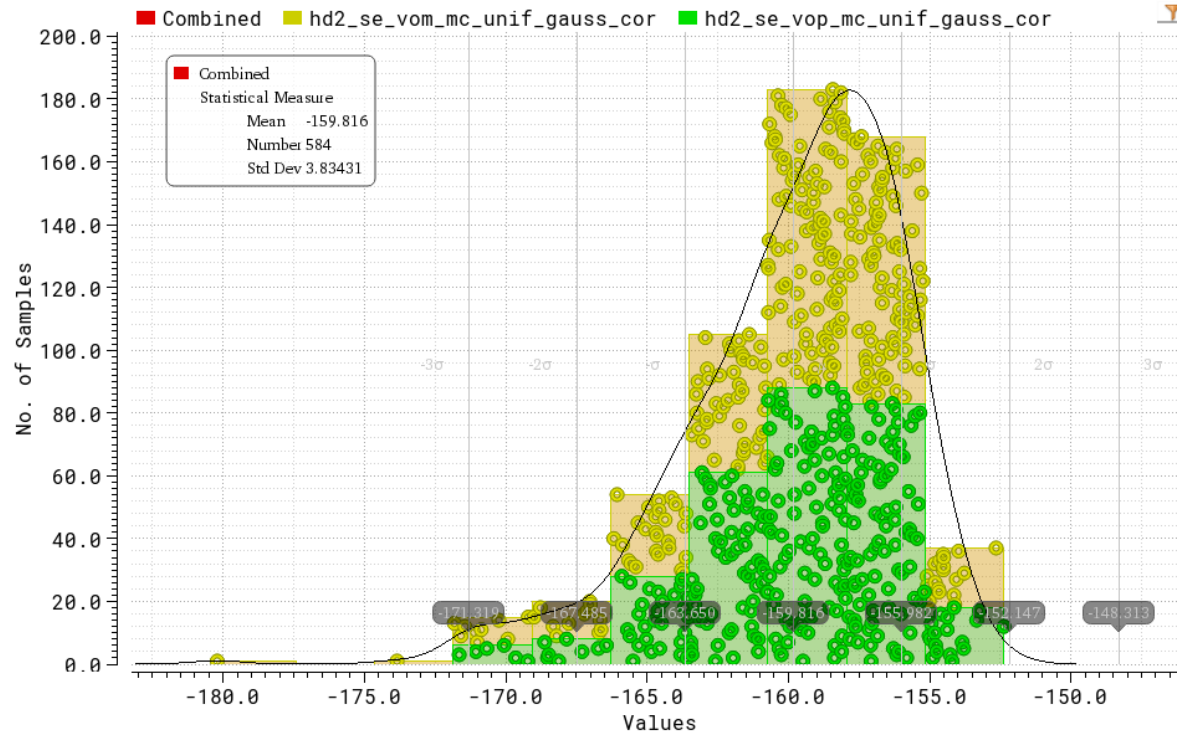
Simulation results of HD2 of the bandpass filter (1kHz). Output amplitude = 10Vppd. Supply $\pm 2.8\text{V}$.

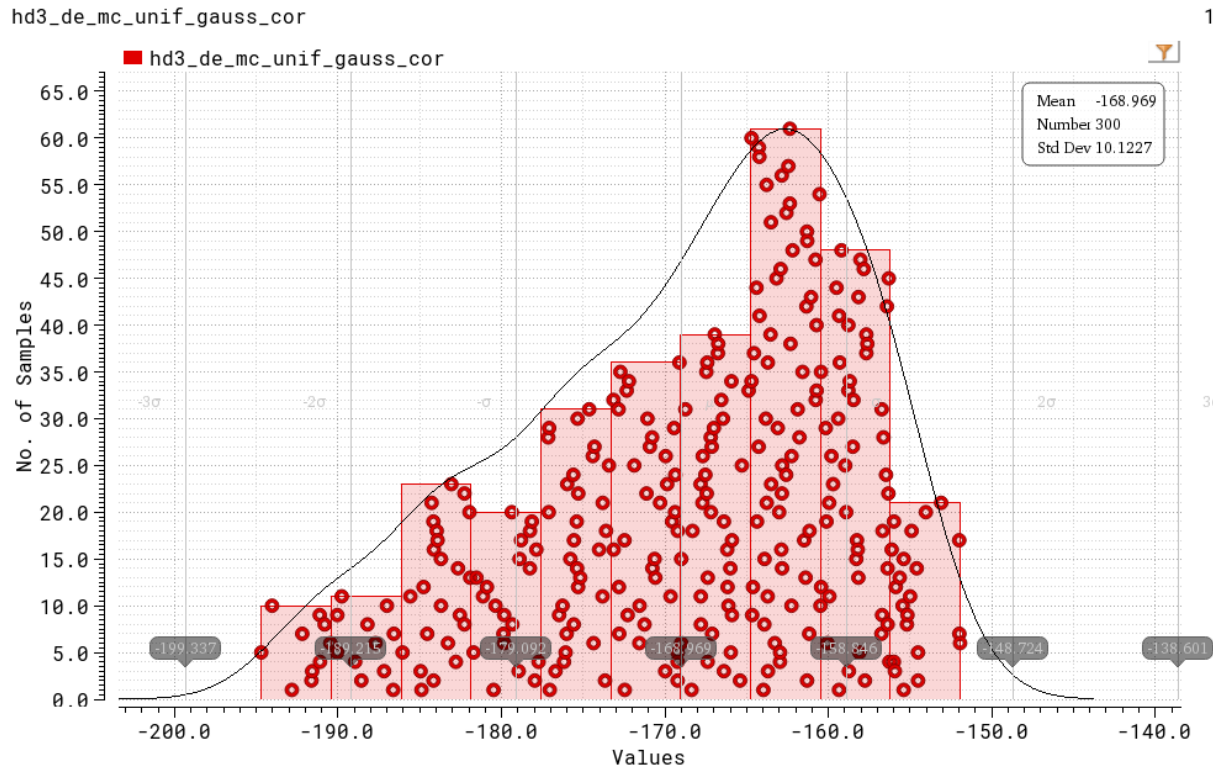
Combined Histogram



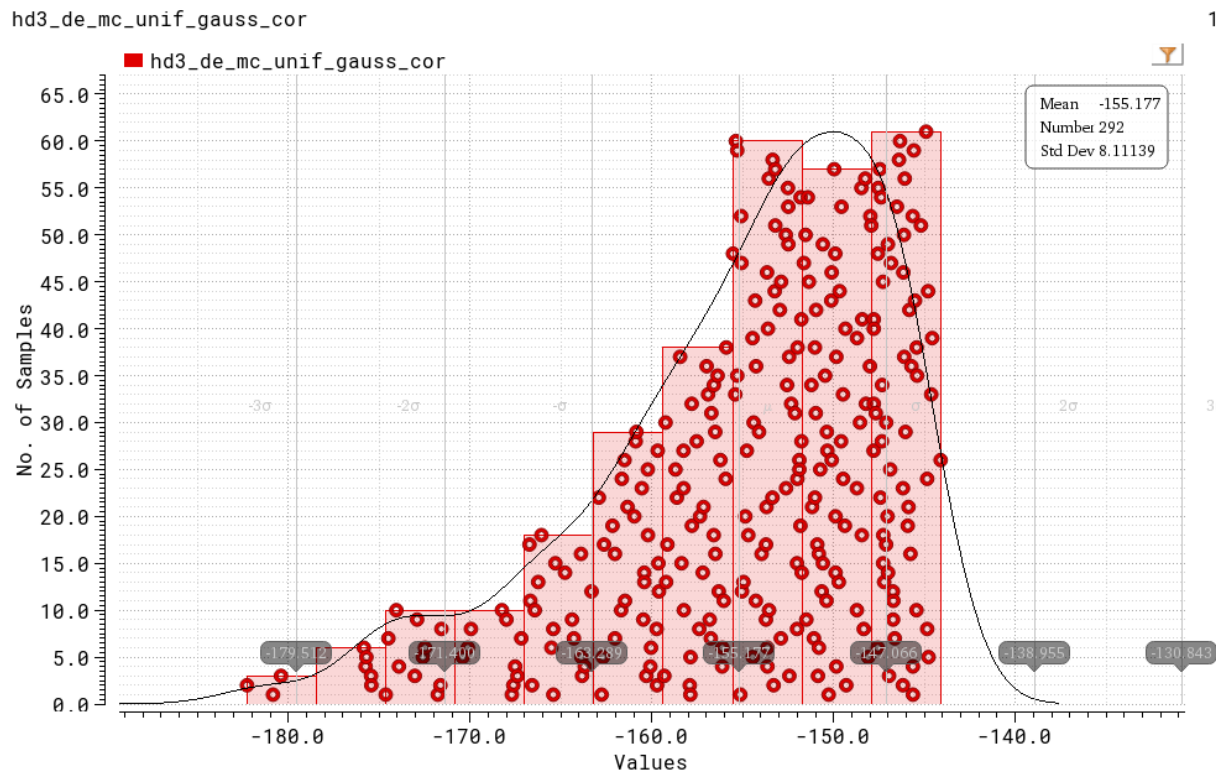
Simulation results of HD2 of the bandpass filter (10kHz). Output amplitude = 10Vppd. Supply $\pm 2.8V$.

Combined Histogram

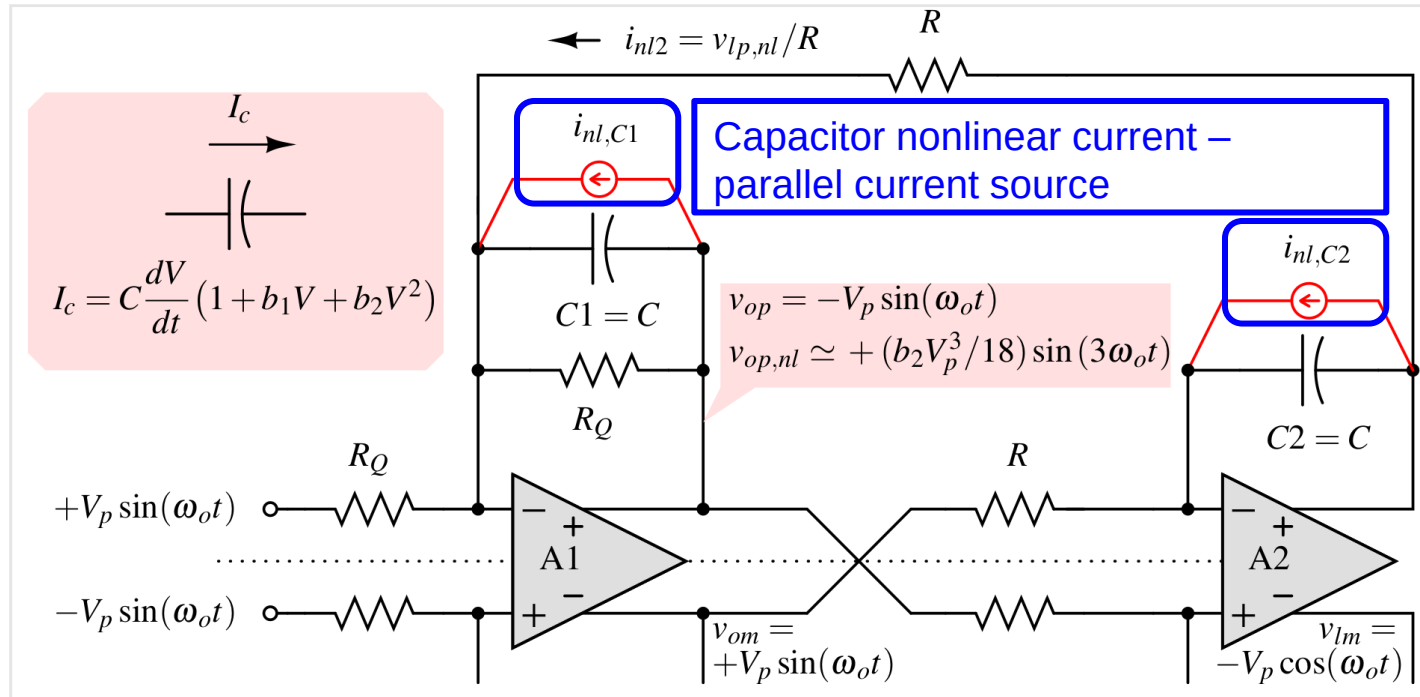




Montecarlo (process and mismatch) results of HD3 of the bandpass filter 10kHz.
Output amplitude = 10Vppd. Supply $\pm 2.8V$



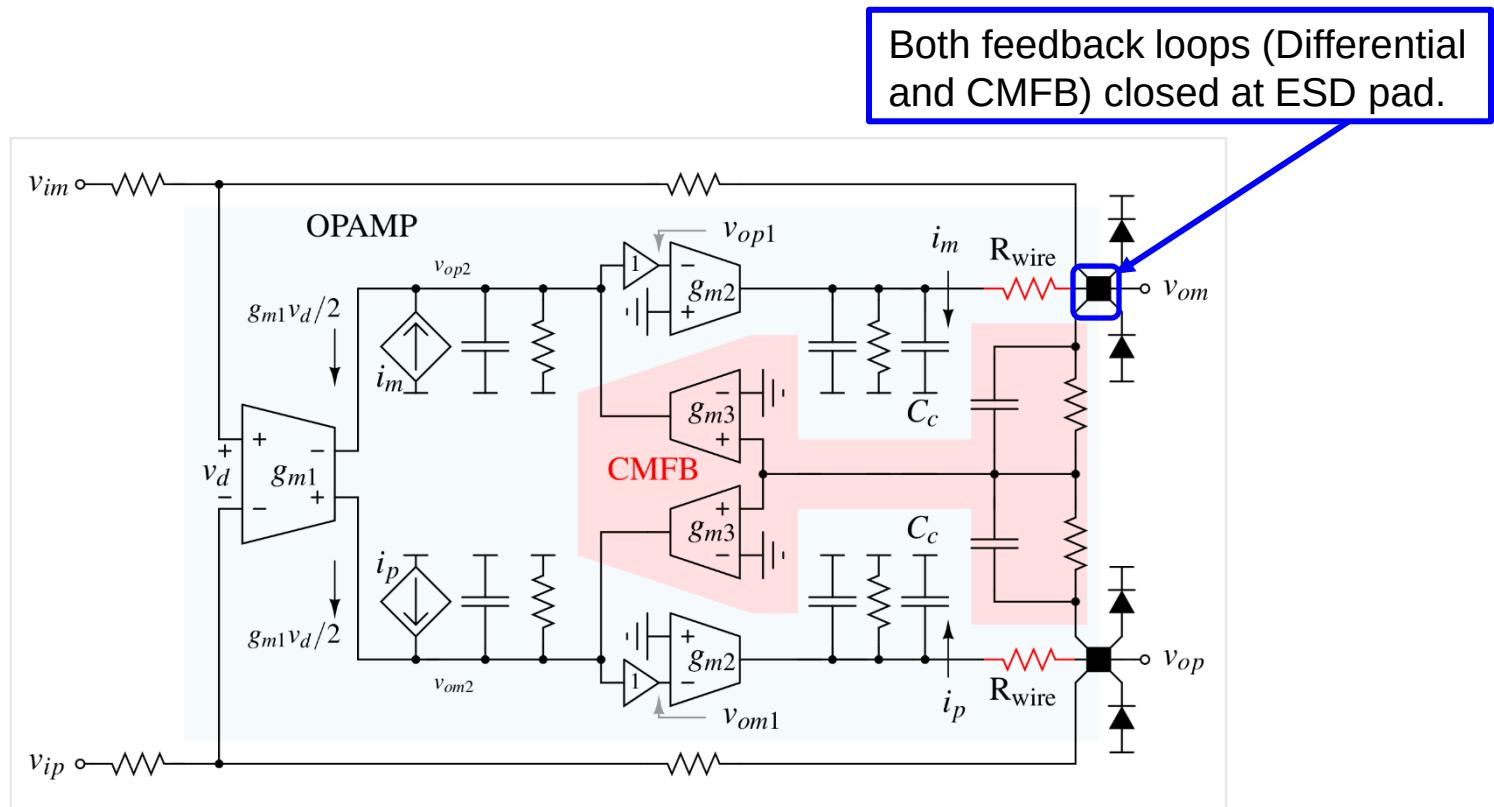
BPF Distortion



$$i_{nl,C1} = + (b_2 V_p^3 / 4R) \cos(3\omega_o t)$$

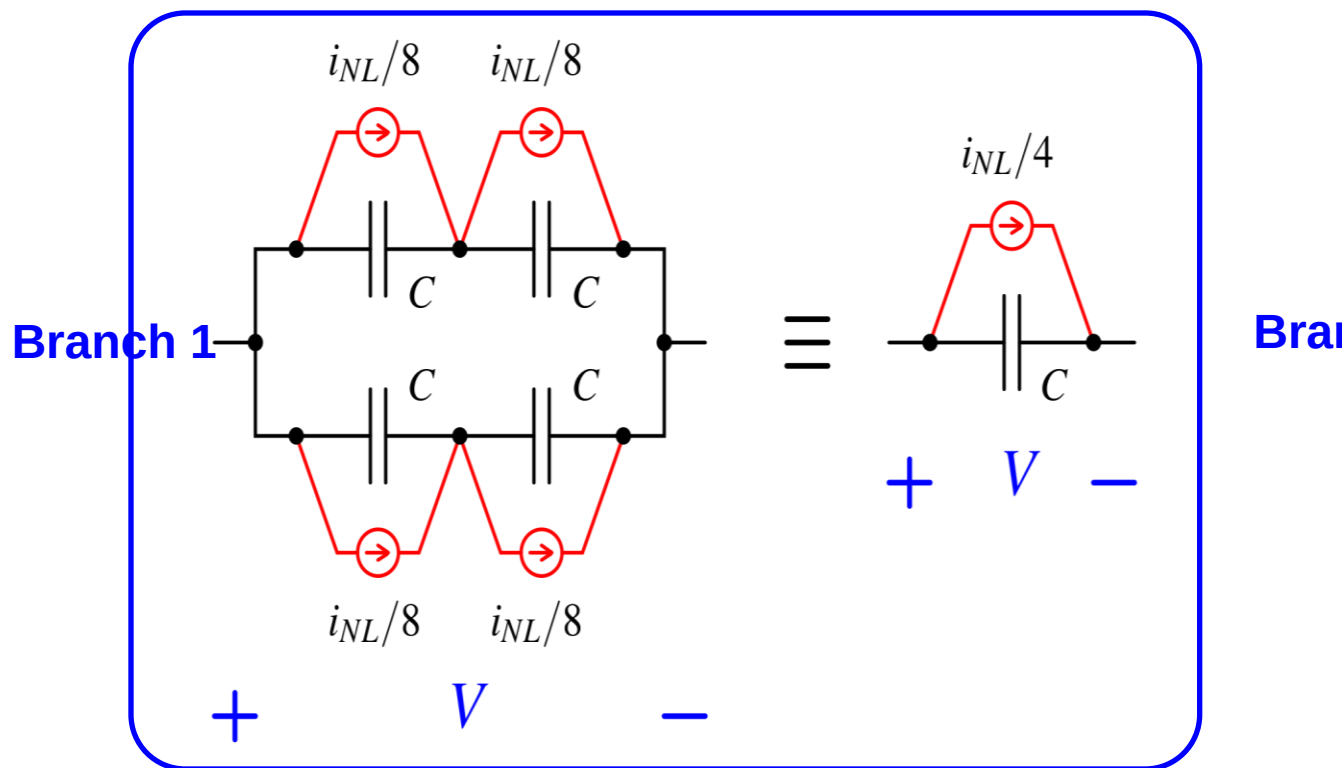
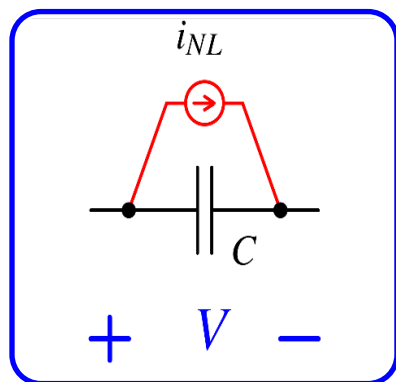
$$i_{nl,C2} = - (b_2 V_p^3 / 4R) \sin(3\omega_o t)$$

ESD's nonlinearity



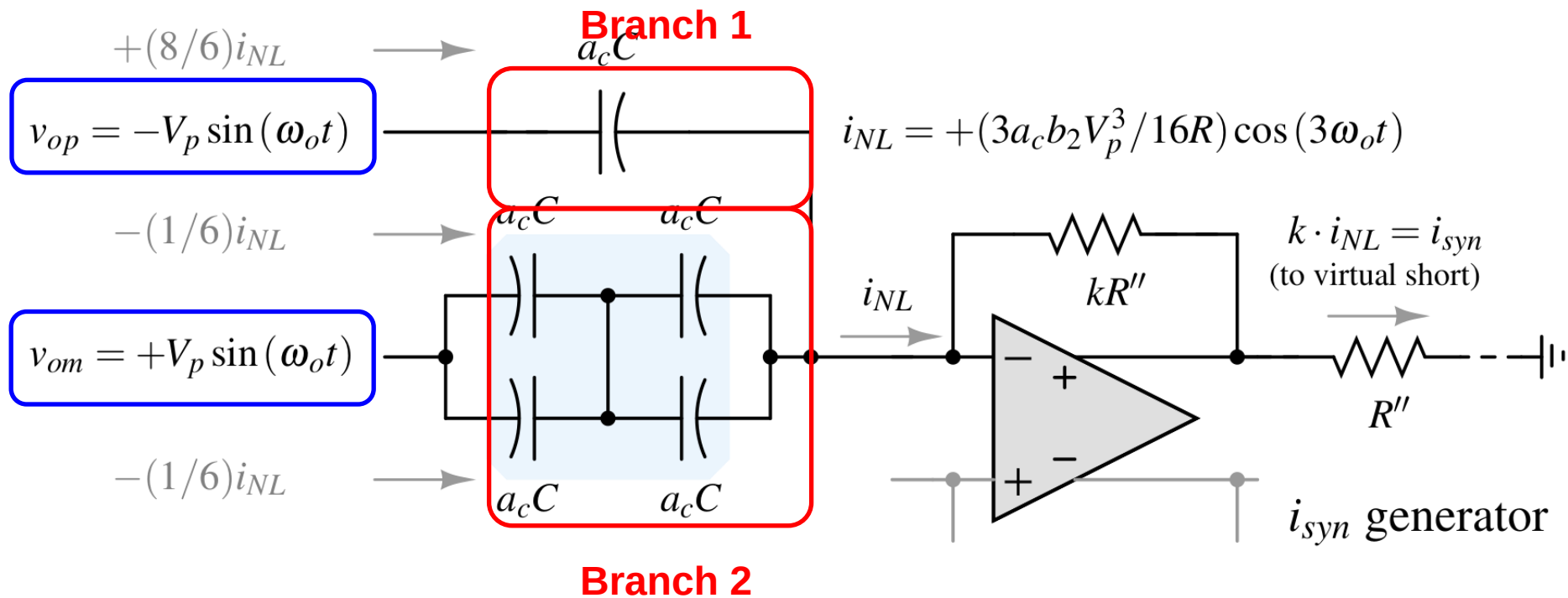
- Star connection suppresses the effect of R_{wire}
- This solves both HD2 and HD3 nonlinearity due to ESD cells

Extracting nonlinear current



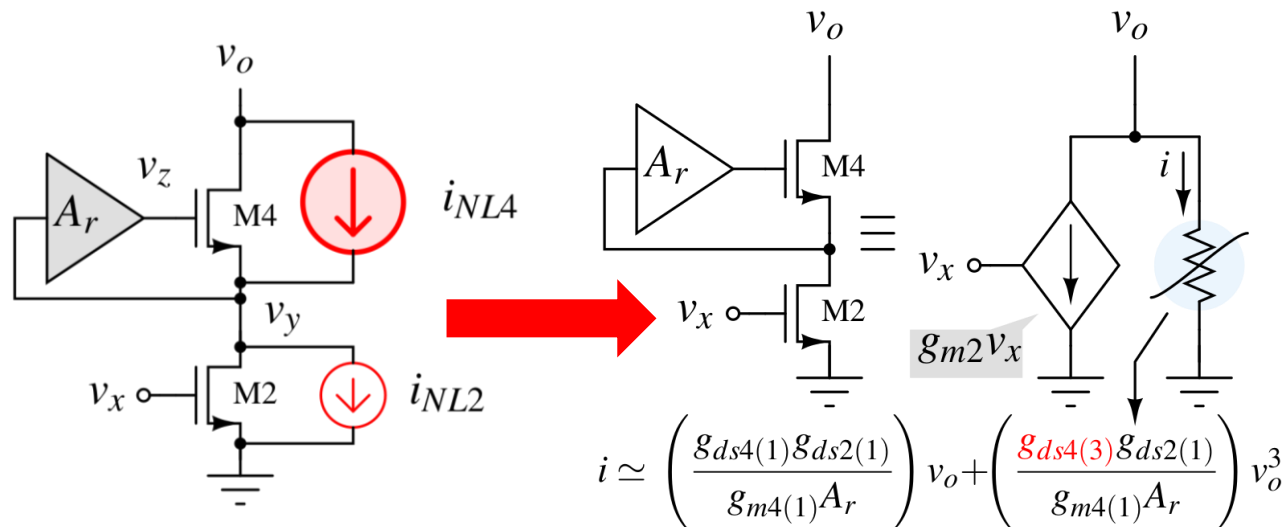
Parallel branches: Identical linear part; Scaled nonlinear part

Extracting nonlinear current



Linear part cancelled; Nonlinear part remains

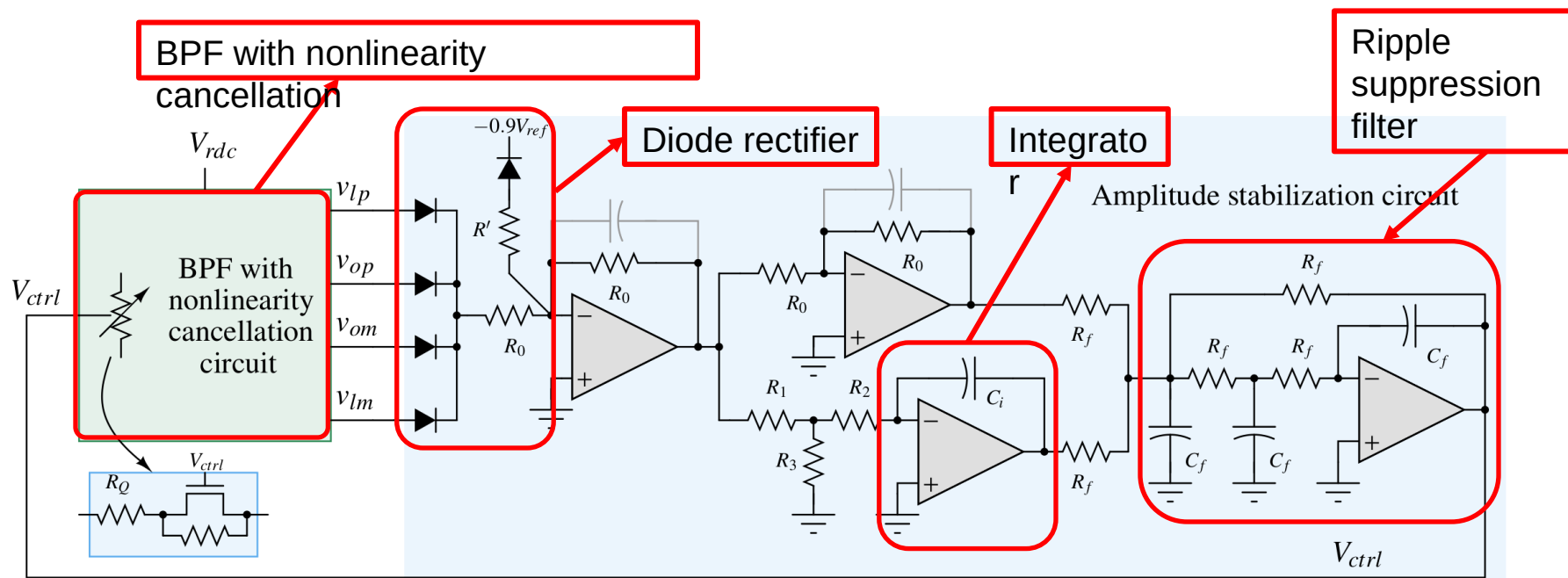
Regulated cascode output stage-more linear



- Small swing at node $v_y \Rightarrow$ negligible i_{NL2}
- Distortion i_{NL4} is suppressed by factor of $A_r g_{m4(1)} / g_{ds2(1)}$

Proposed Solution – Approach 2

Oscillator (BPF + Amplitude stabilization)

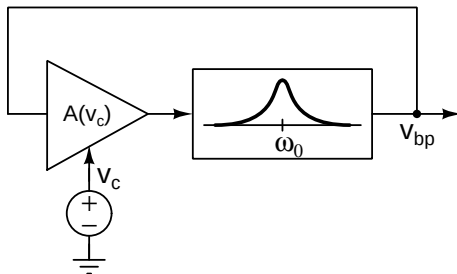


Low-distortion Oscillators

- Sinusoidal oscillator
- Amplitude stabilization
- Oscillator distortion
- Measurement results
- Comparison and conclusion

Oscillator core loop

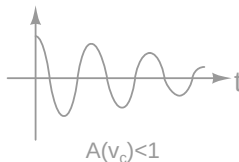
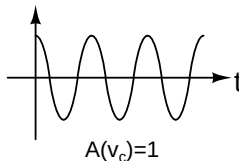
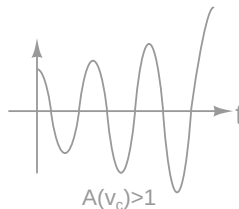
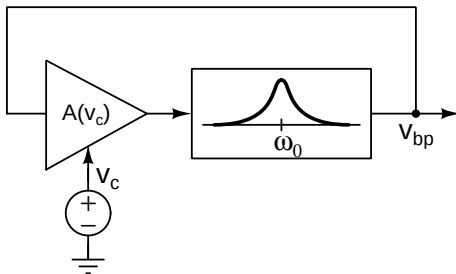
- Band-pass filter in feedback oscillates depending on loop gain



Effect of loop gain

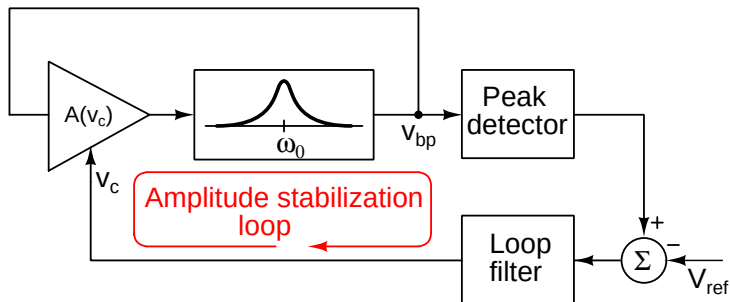
- Band-pass filter in feedback oscillates depending on loop gain
- $A(v_c) > 1$, oscillations build up until saturation
- If $A(v_c) < 1$, oscillations die out
- $A(v_c) = 1$, produces stable oscillations

$$v_{bp} = V_{p0} \cos(\omega_0 t)$$

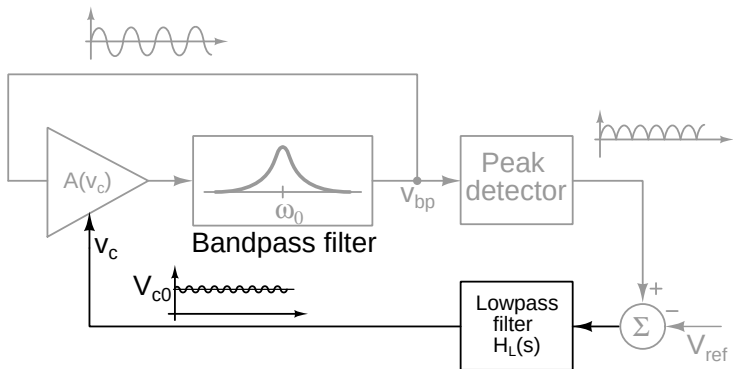


Amplitude stabilization loop

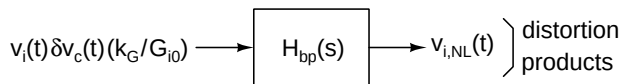
- Voltage v_c needs to be controlled using a negative feedback loop



- Peak detector has output ripple
- Low-pass filter $H_L(s)$ is needed to attenuate ripple



- Opamp: Designed for low distortion
- Peak detector ripple:
 - Four phase peak detector
 - Additional pole in $H_L(s)$
 - Low dA/dv_C ; But certain range of A is required for guaranteed oscillation without saturation across corners

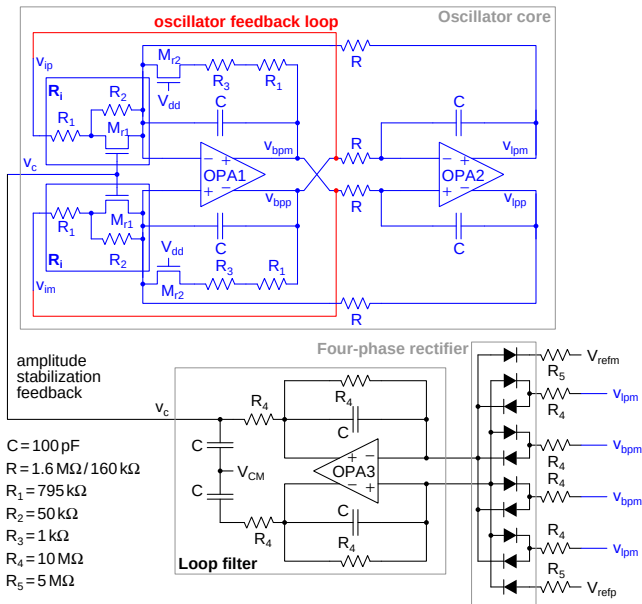


$$HD_3 = \underbrace{(4/15\pi)}_{-121dB} \underbrace{|H_{cr}(j4\omega_o)|}_{-21dB} \underbrace{(k_G V_{p0}/G_{i0})}_{-55dB} \underbrace{|H_{bp}(j3\omega_o)|}_{-45dB} \underbrace{|H_{bp}(j3\omega_o)|}_{\approx 0dB}$$

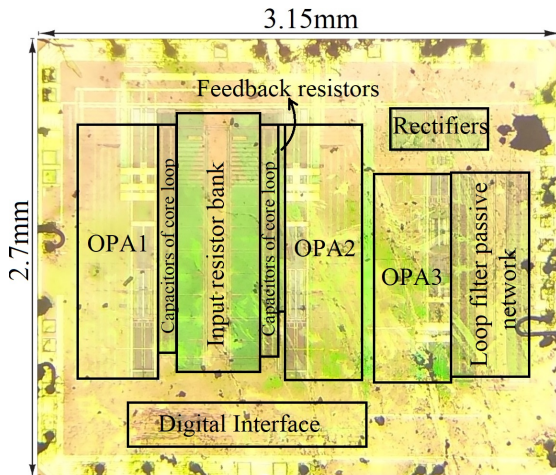
- 2 poles in $H_L(s)$ at 160Hz

Oscillator implementation

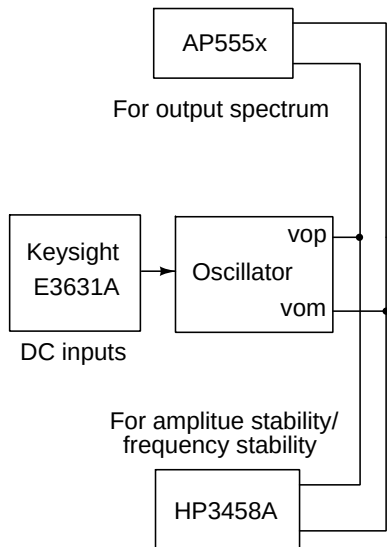
Diodes in reference path compensate the variations in rectifier characteristics



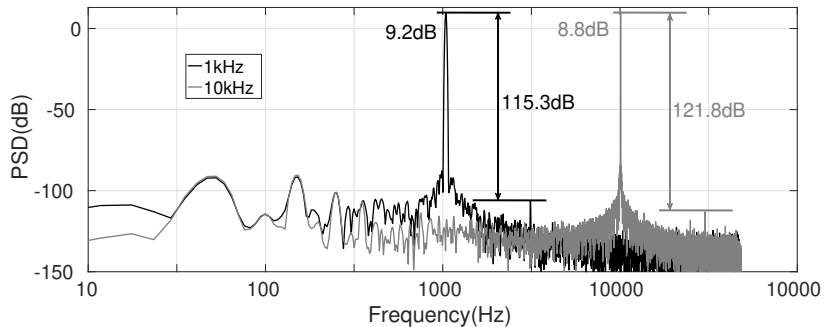
- Differential symmetry is maintained between all the passive components
- Low common routing resistance between rectifier input and output pads
- Feedback resistors in the oscillator loop kept close to the opamp to reduce the parasitics



Oscillator test setup

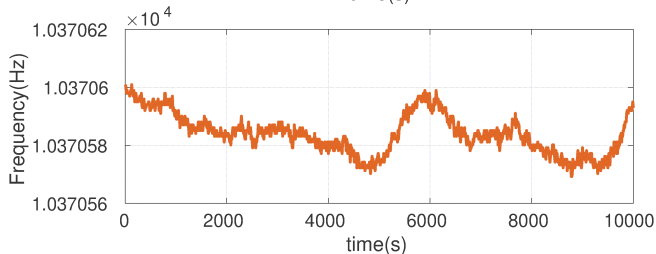
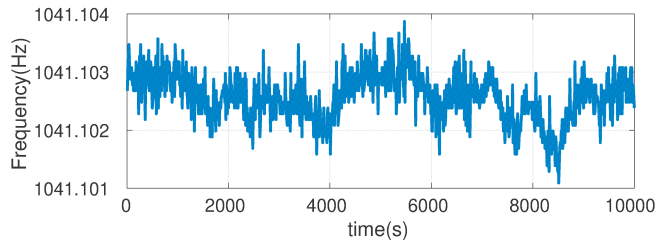


Oscillator: Measured distortion



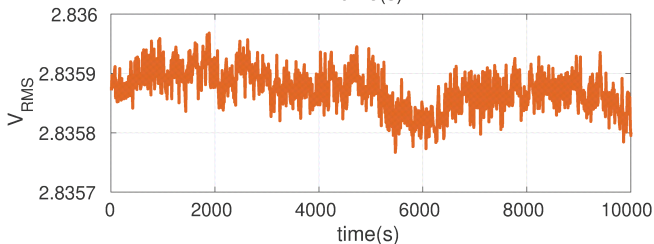
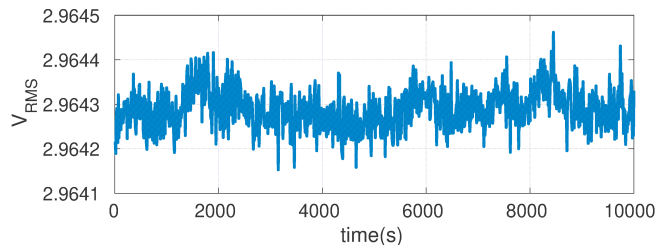
Frequency (kHz)	Amplitude(V_{pd})	THD(dB)
1	4.07	-115.3
10	3.89	-121.8

Oscillator: Measured frequency stability



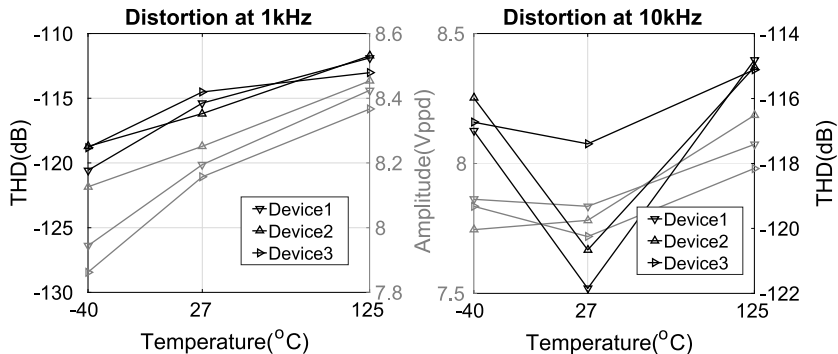
f_o (kHz)	$\sigma(f_o)$ (mHz)
1	0.75
10	13

Oscillator: Measured amplitude stability



f_o (kHz)	$\sigma(A_o)$ (μ V)
1	82
10	43.1

Measured THD for 3 devices



- THD < -112 dB at 1 kHz and < -115 dB at 10 kHz

Performance summary and comparison

	This work	Vasan TCAS1, 2013	APP555x	Elsayed JSSC, 2010,	Shi TCAS1, 2015	Shi JSSC, 2017
Process	0.6 μm	Discrete	Bench-top	0.13 μm	0.18 μm	0.13 μm
Area(mm^2)	8.5			0.1(SE) 0.19(Diff)	0.08	0.056
Power(mW)	49.5			3.34(SE) 4.04(Diff)	9.1–57	6
THD(dB)	-114.8 (1kHz) -116.5 (10 kHz)	-82 (29kHz) -100 (2.7 kHz)	<-117 (1kHz)	-57.9 (SE) -72 (Diff)	-54.9 (150MHz) -57.7 (650 MHz) -62.6	-66 (720MHz) -68 (480 MHz) -80 (20 MHz)
Frequency (Hz)	1k, 10k	29k, 2.7k	5–204k	10M	150M–850M	< 1G
Supply (V)	5	30		1.2	1–1.8	1.2/1.5
Vpp (V)	8	7/1.55		0.15		

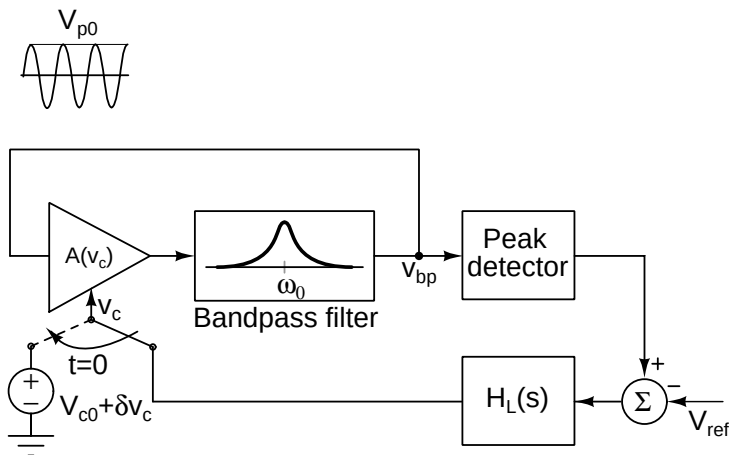
- Low-distortion filter and oscillator ICs for ADC testing in 0.6 μm CMOS
- Reducing opamp distortion
 - Reduce the nonlinear current driven by the first stage
 - Make the second stage more linear using a push-pull topology
 - Reduce output conductance nonlinearity by using a regulated cascode
- Reducing passive component distortion
 - Cancel the distortion due to capacitors using a replica to extract the third harmonic
 - Use high-linearity metal-film resistors
- Reducing distortion due to output wiring and nonlinear ESD diodes
 - Four-wire connection between opamp/feedback network and output pads
- Peak-detector ripple additionally contributes to oscillator distortion
- Filter with THD better than -140 dB demonstrated
- Oscillator with THD better than -130 dB demonstrated

Getting to -140 dBc THD

Backup slides

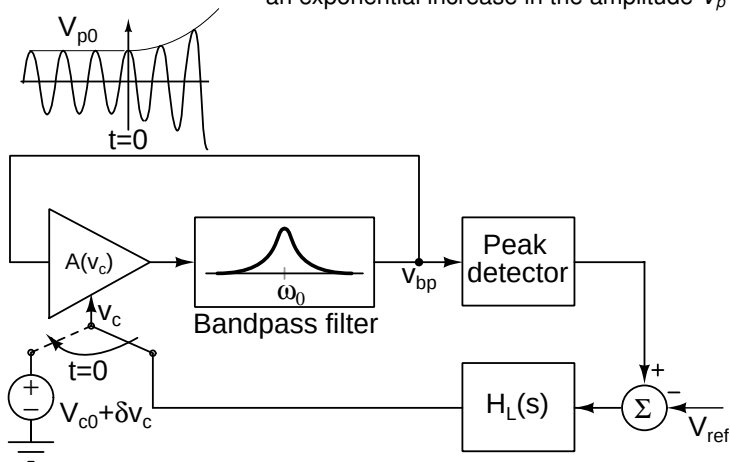
Amplitude stabilization loop modeling

- If $v_c = V_{c0}$, then $v_{bp} = V_{p0} \cos \omega_0 t$



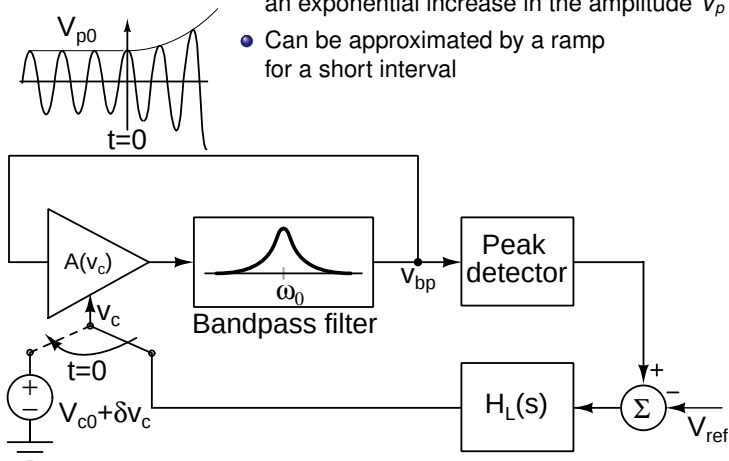
Amplitude stabilization loop modeling

- If $v_c = V_{c0}$, then $v_{bp} = V_{p0} \cos \omega_o t$
- Step change δv_c in v_c causes an exponential increase in the amplitude V_p



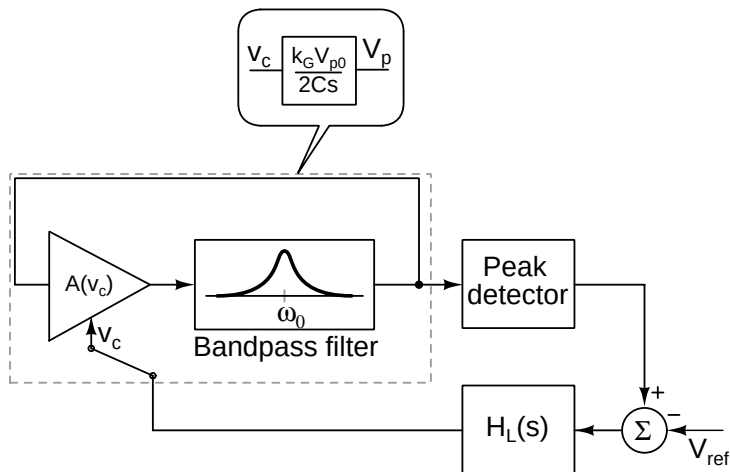
Amplitude stabilization loop modeling

- If $v_c = V_{c0}$, then $v_{bp} = V_{p0} \cos \omega_0 t$
- Step change δv_c in v_c causes an exponential increase in the amplitude V_p
- Can be approximated by a ramp for a short interval

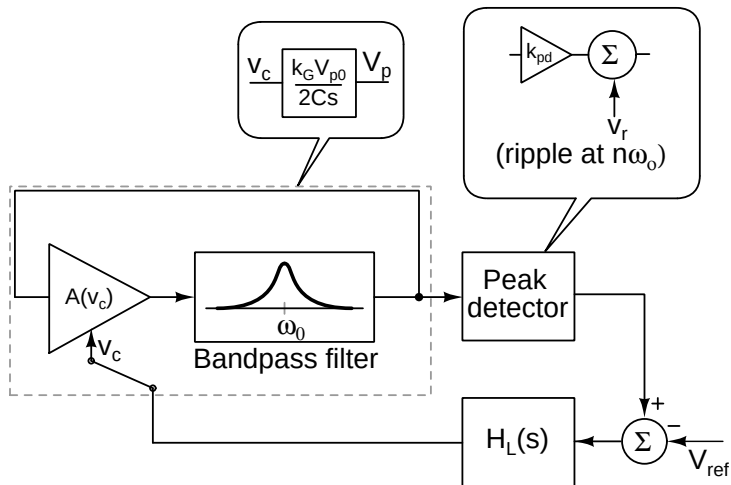


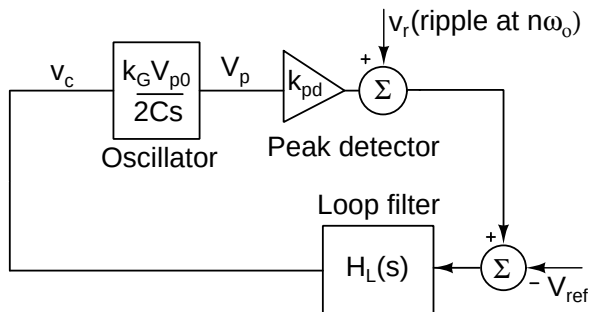
Amplitude stabilization loop modeling

- Oscillator $\approx$ integrator from v_c to output amplitude



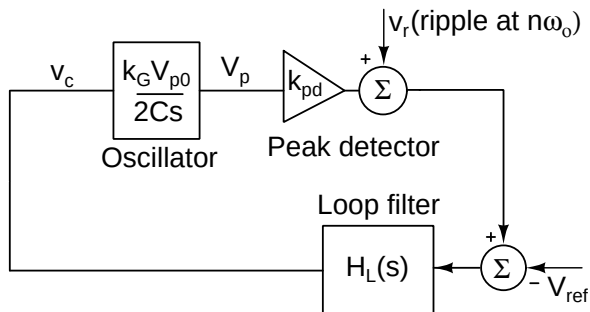
Amplitude stabilization loop modeling





- Ripple in the control voltage

$$V_c(s) = H_{cr}(s) V_r(s) = \frac{H_L(s)}{1 + (k_G V_{p0} / 2Cs) k_{pd} H_L(s)} V_r(s)$$



- Ripple in the control voltage

$$V_c(s) = H_{cr}(s) V_r(s) = \frac{H_L(s)}{1 + (k_G V_{p0} / 2Cs) k_{pd} H_L(s)} V_r(s)$$

- Four phase rectifier reduces ripple amplitude, increases ripple frequency to $4\omega_o$